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(54) **PHOTOELECTRIC CONVERSION APPARATUS, PHOTOELECTRIC CONVERSION SYSTEM, MOVING BODY, AND EQUIPMENT**

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(57) **ABSTRACT**

A back-illuminated type photoelectric conversion apparatus in which a first semiconductor layer including an APD array in which a plurality of avalanche photodiodes (APDs) are arranged, a second semiconductor layer including a plurality of first pixel circuits respectively for the plurality of APDs, a third semiconductor layer including a plurality of second pixel circuits respectively for the plurality of first pixel circuits, and a fourth semiconductor layer including a processing circuit configured to process signals output from the plurality of second pixel circuits are stacked is provided. The second semiconductor layer is arranged between the first semiconductor layer and the fourth semiconductor layer, and the third semiconductor layer is arranged between the second semiconductor layer and the fourth semiconductor layer. A through via extending through the third semiconductor layer is arranged in a region not overlapping the APD array.

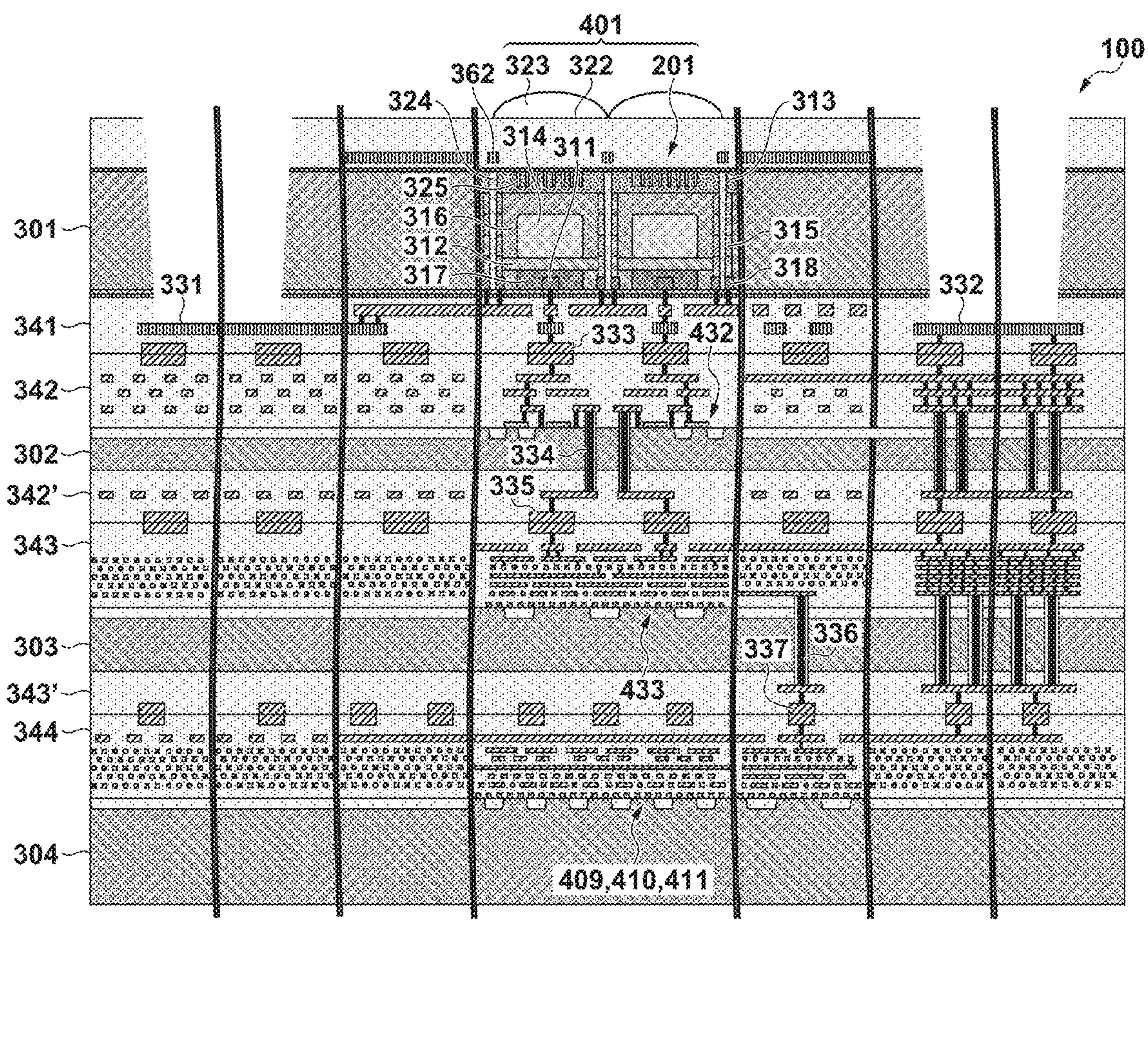


FIG. 1

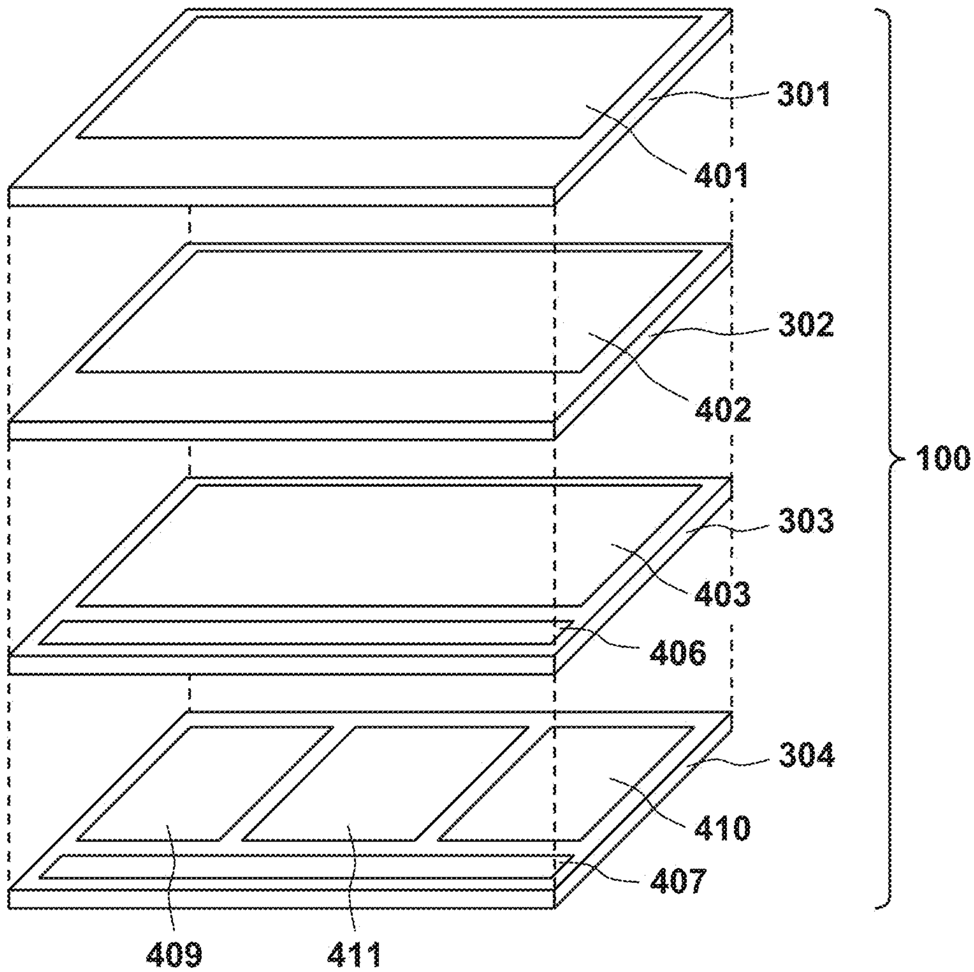
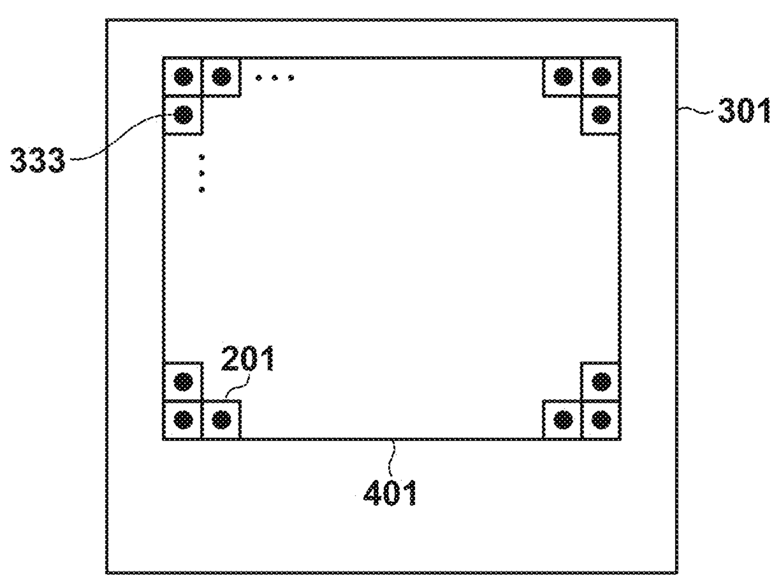
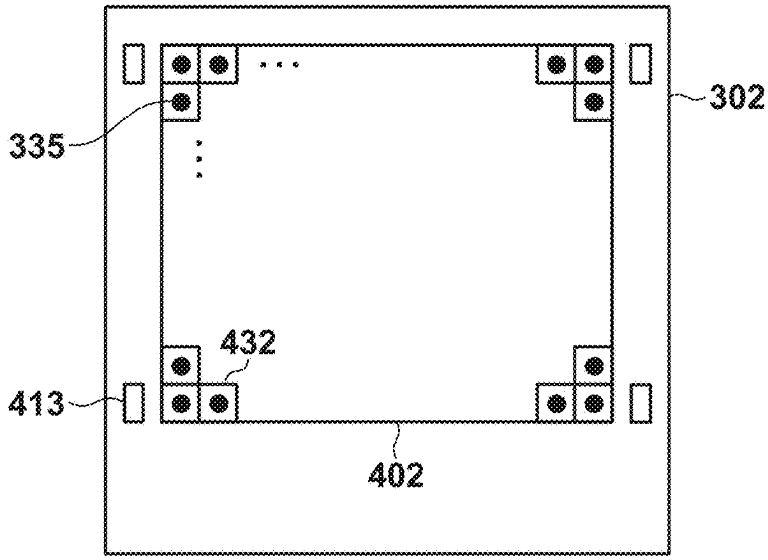

301
401
302
402
100
303
403
406
304
410
407
409
411

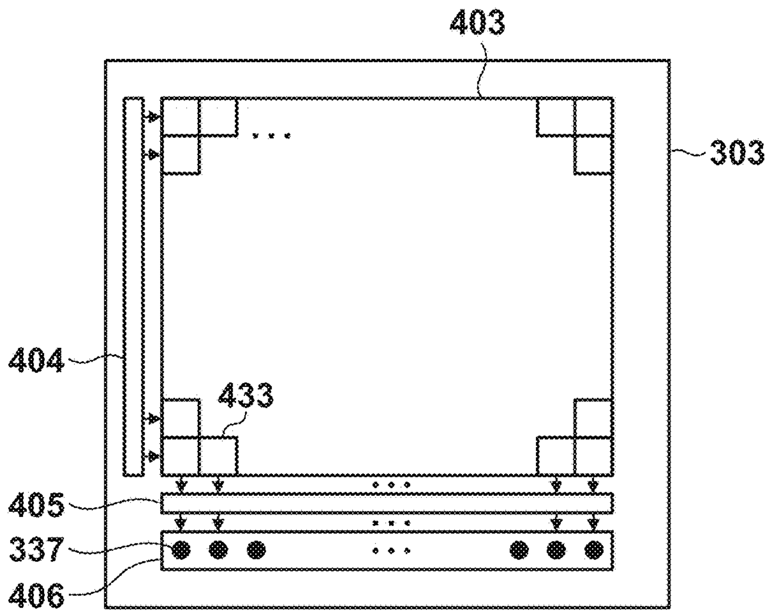
FIG. 2A


301
333
201
401

F I G. 2B



F I G. 2C



F I G. 2D

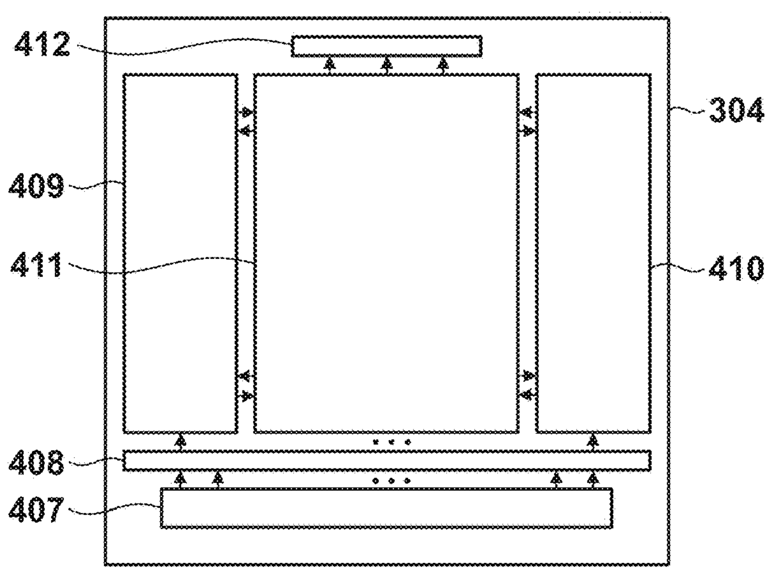


FIG. 3

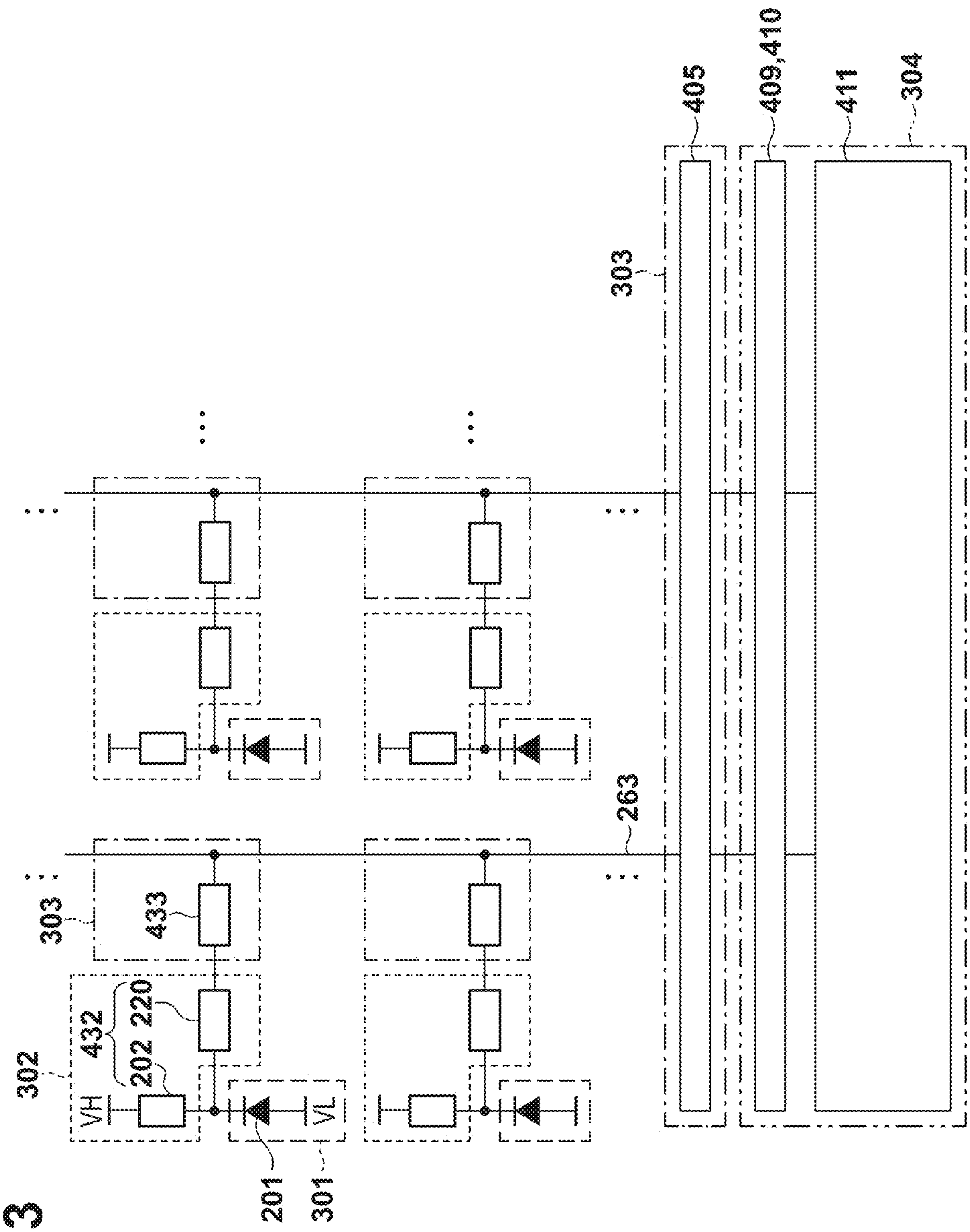


FIG. 4A

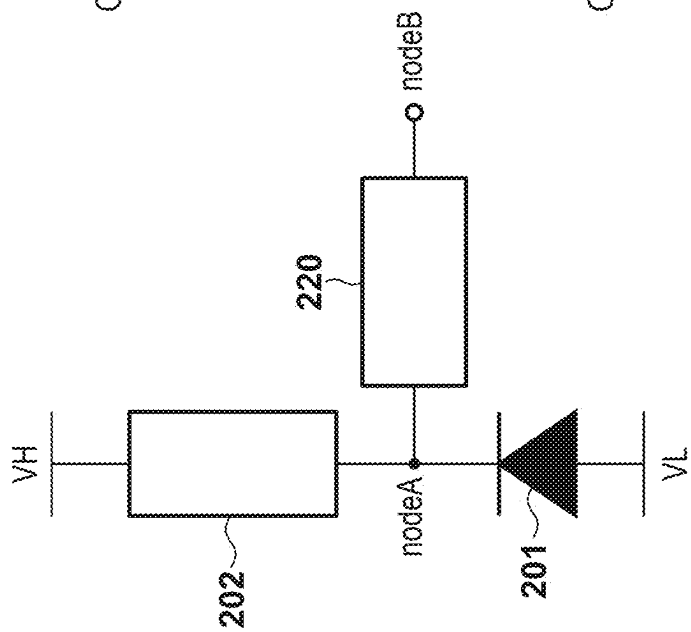

VH
202
nodeA
220
nodeB
201
VL

FIG. 4B

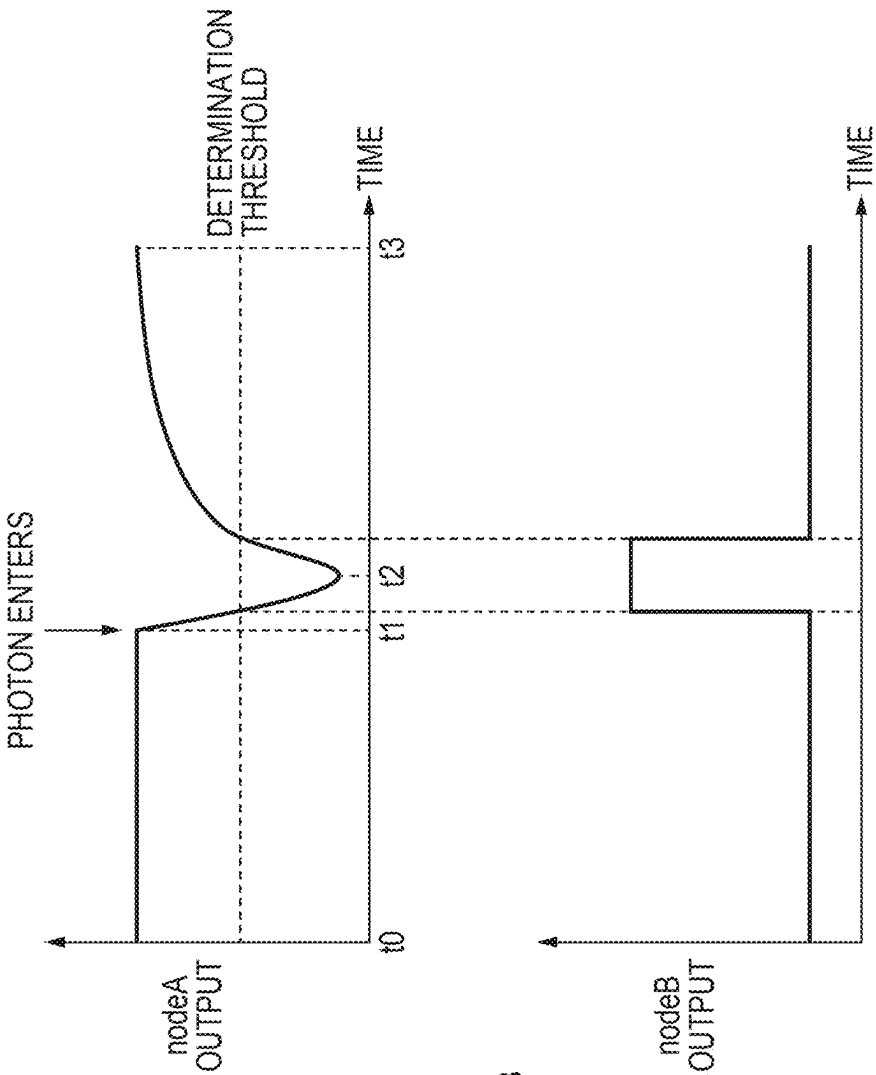

PHOTON ENTERS
DETERMINATION
THRESHOLD
nodeA
OUTPUT
TIME
t0
t1
t2
t3
nodeB
OUTPUT
TIME

FIG. 5A

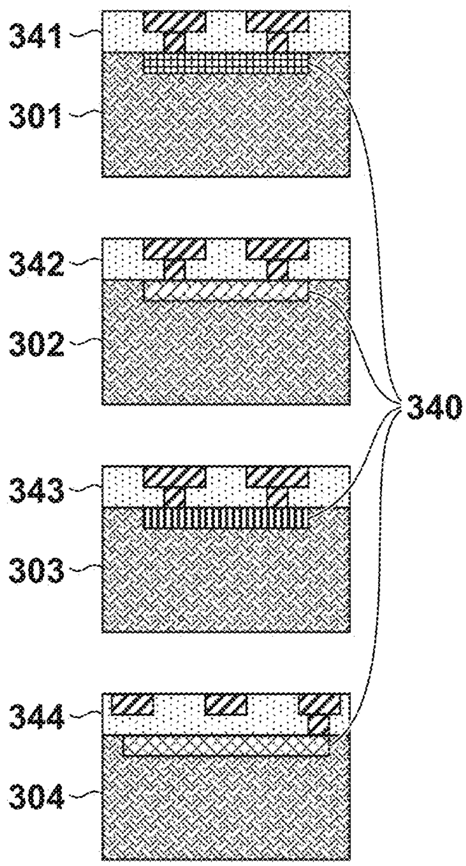


FIG. 5B

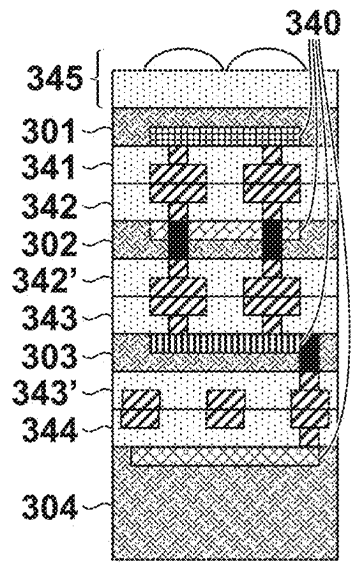


FIG. 5C

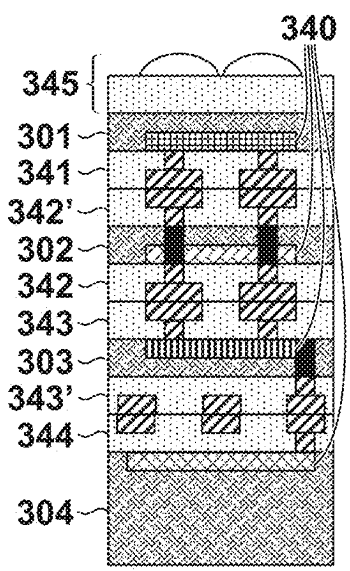


FIG. 5D

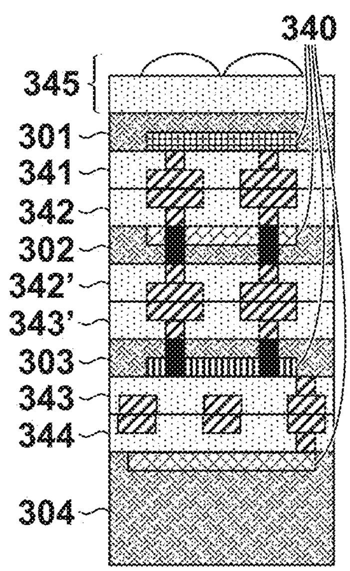


FIG. 5E

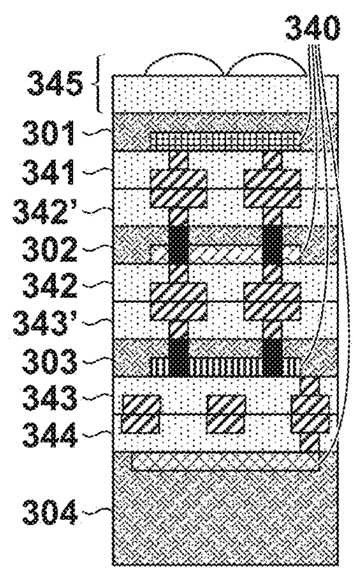


FIG. 5F

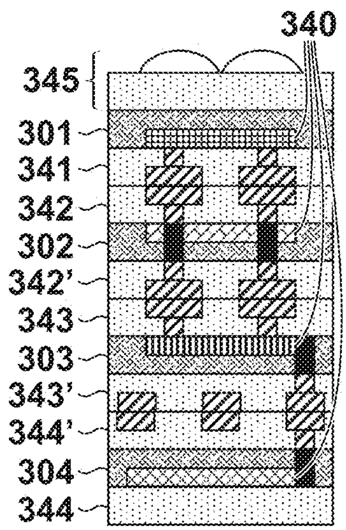


FIG. 5G

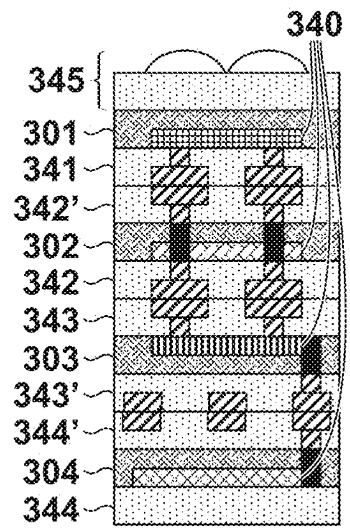


FIG. 5H

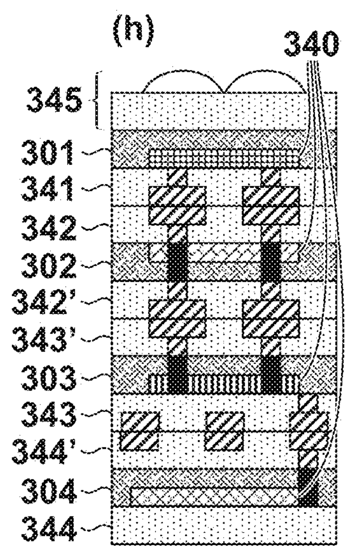
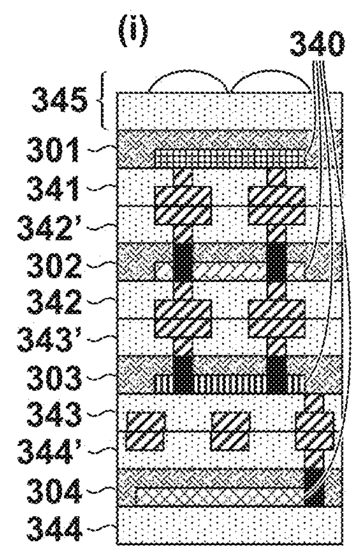


FIG. 5I



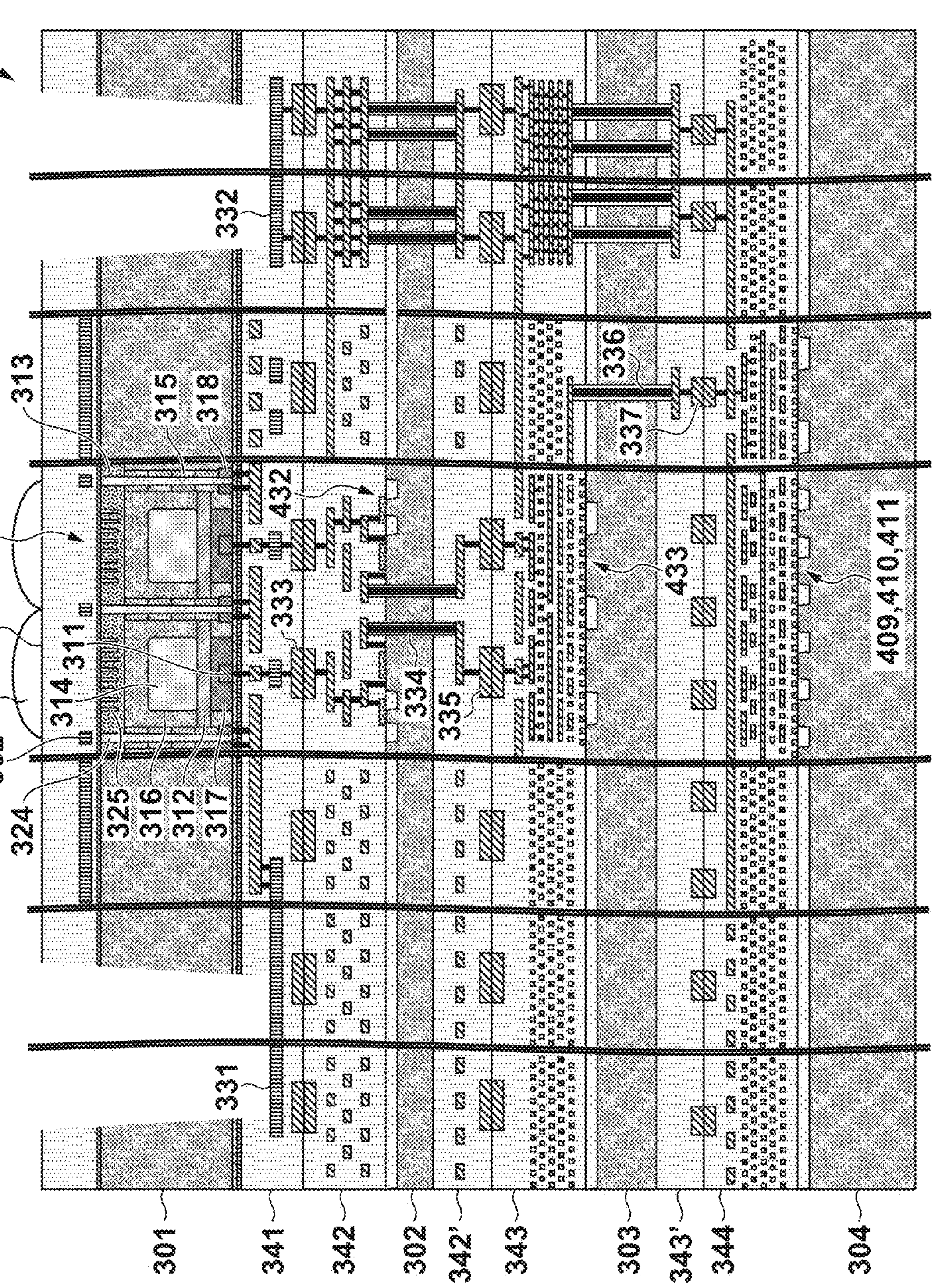
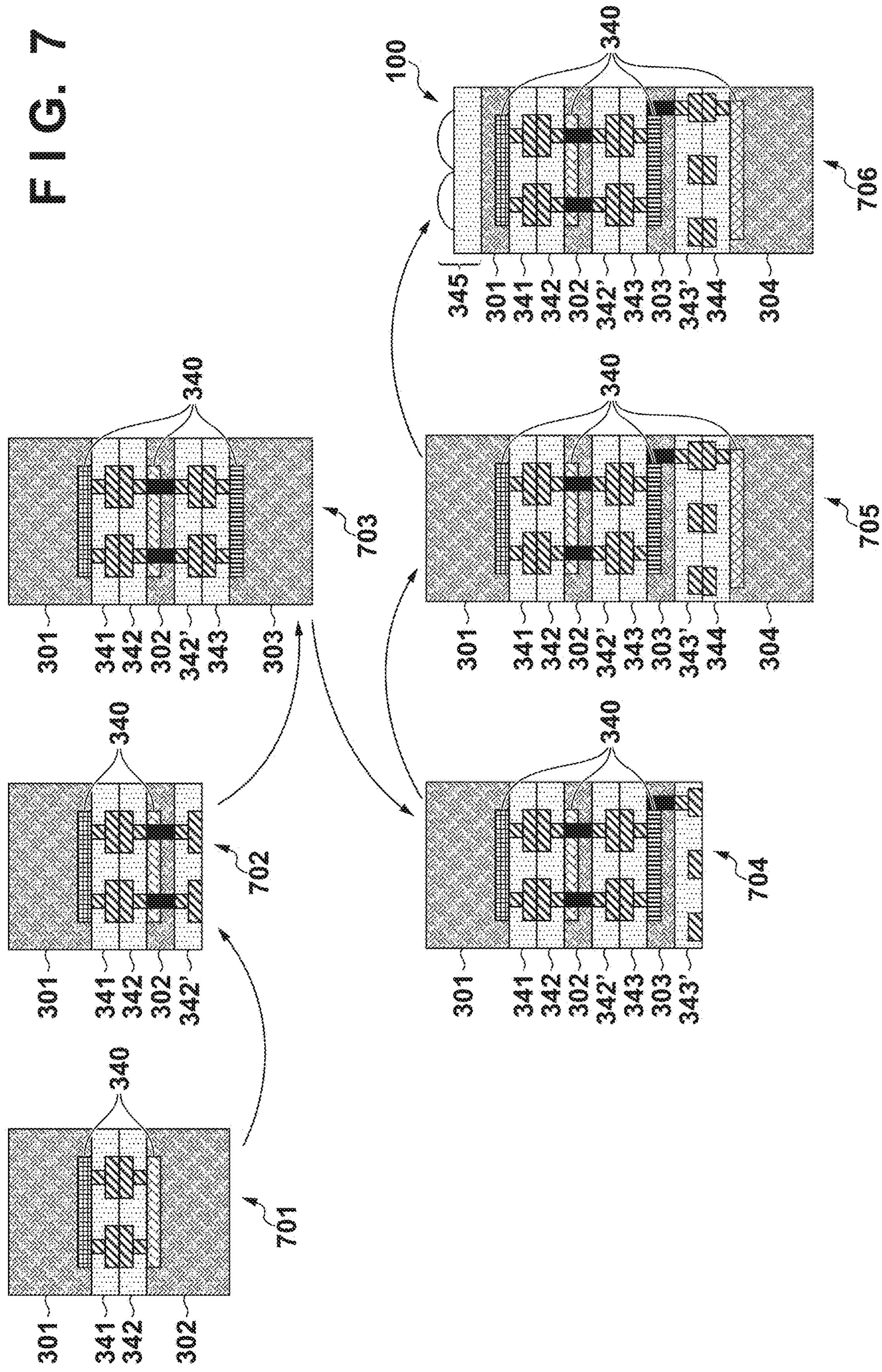

100
332
313
315
318
336
337
401
201
322
323
311
314
362
333
432
433
334
335
409,410,411
324
325
316
312
317
331
301
341
342
302
342'
343
303
343'
344
304

FIG. 6

FIG. 7



701
702
703
704
705
706
100
340
301
302
303
304
341
342
342'
343
343'
344
345

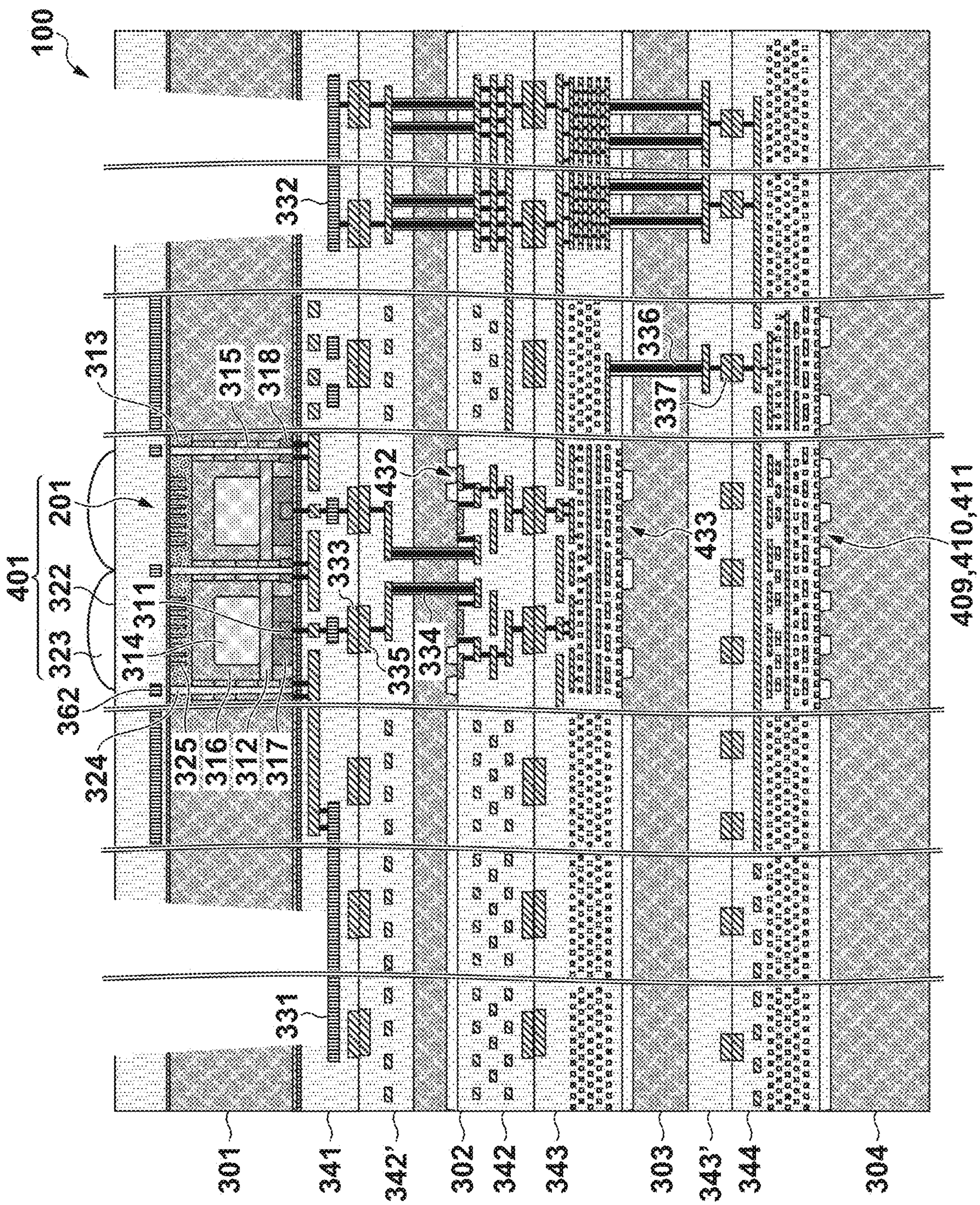
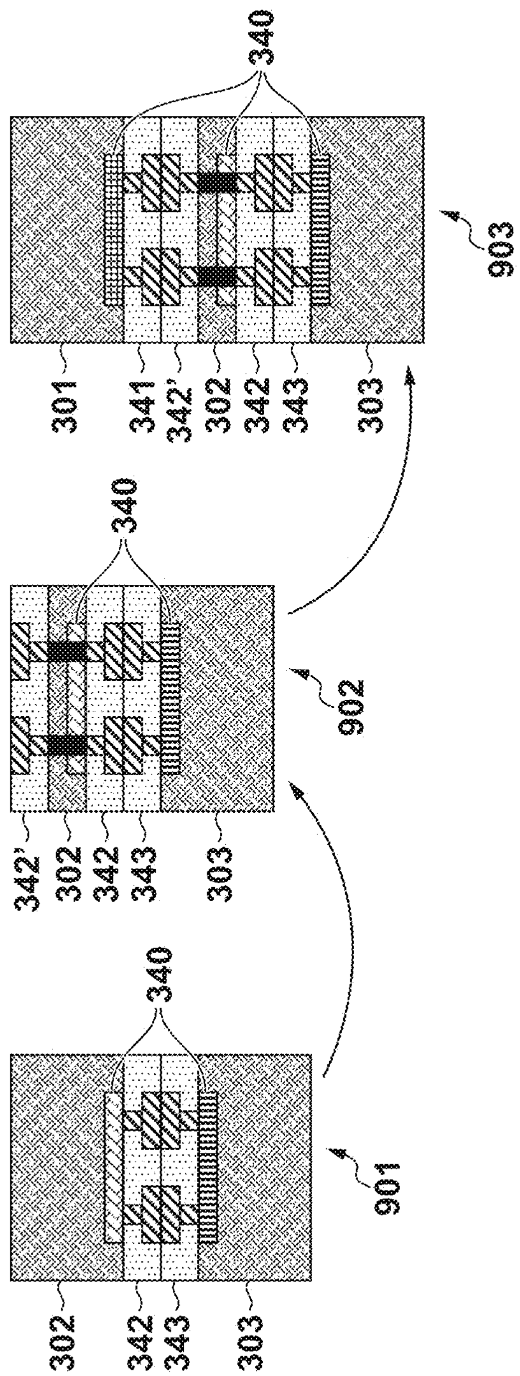
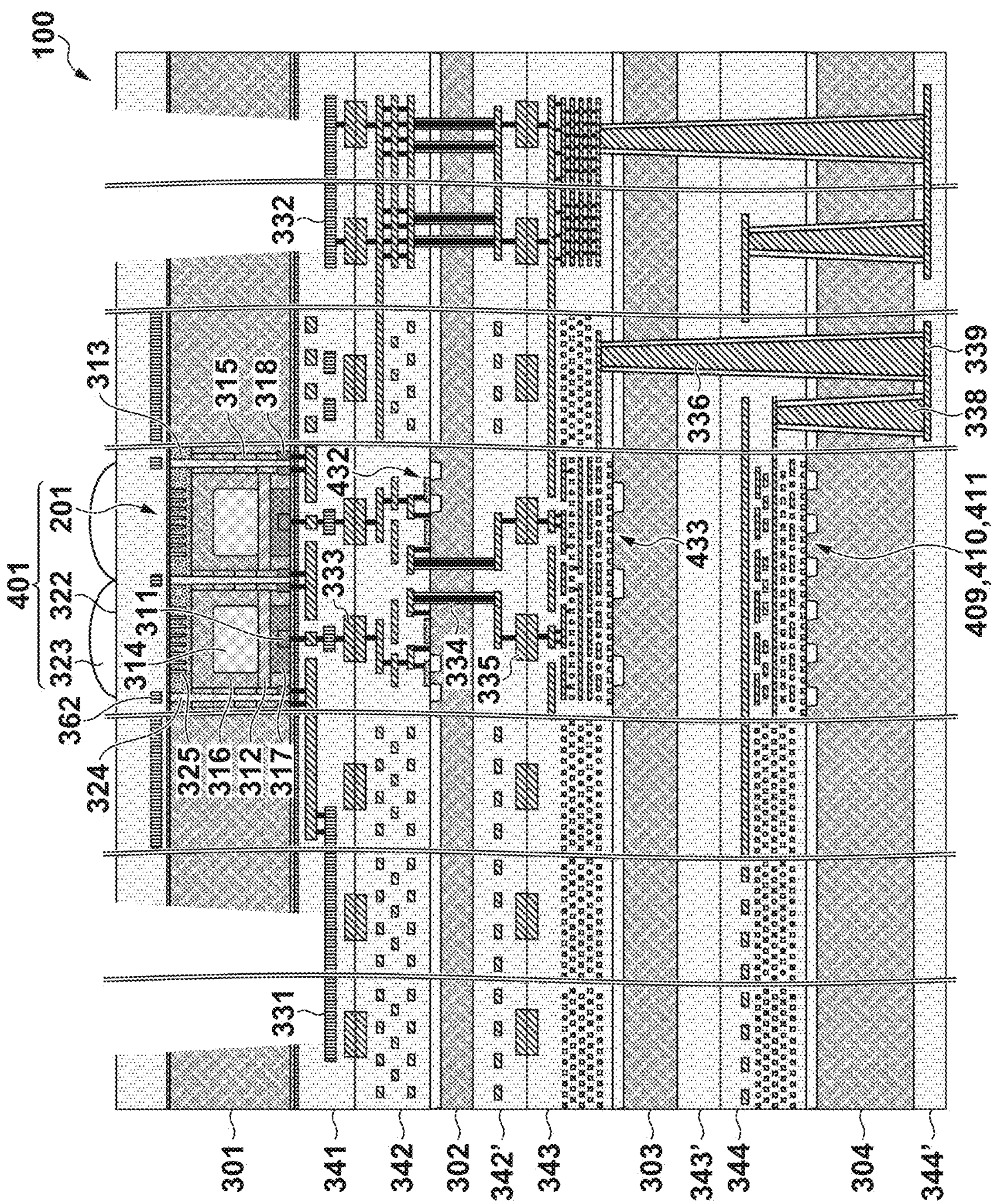

100
332
313
315
318
401
322
201
323
311
333
432
362
314
335
334
324
325
316
312
317
336
337
433
409,410,411
331
301
341
342'
302
342
343
303
343'
344
304

FIG. 8

FIG. 9



340
301
341
342'
302
342
343
303
903
340
342'
302
342
343
303
902
340
302
342
343
303
901


100
401
201
322
323
311
313
314
315
318
432
333
334
335
362
324
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316
312
317
331
332
336
338
339
433
409,410,411
301
341
342
302
342'
343
303
343'
344
304
344'

FIG. 10

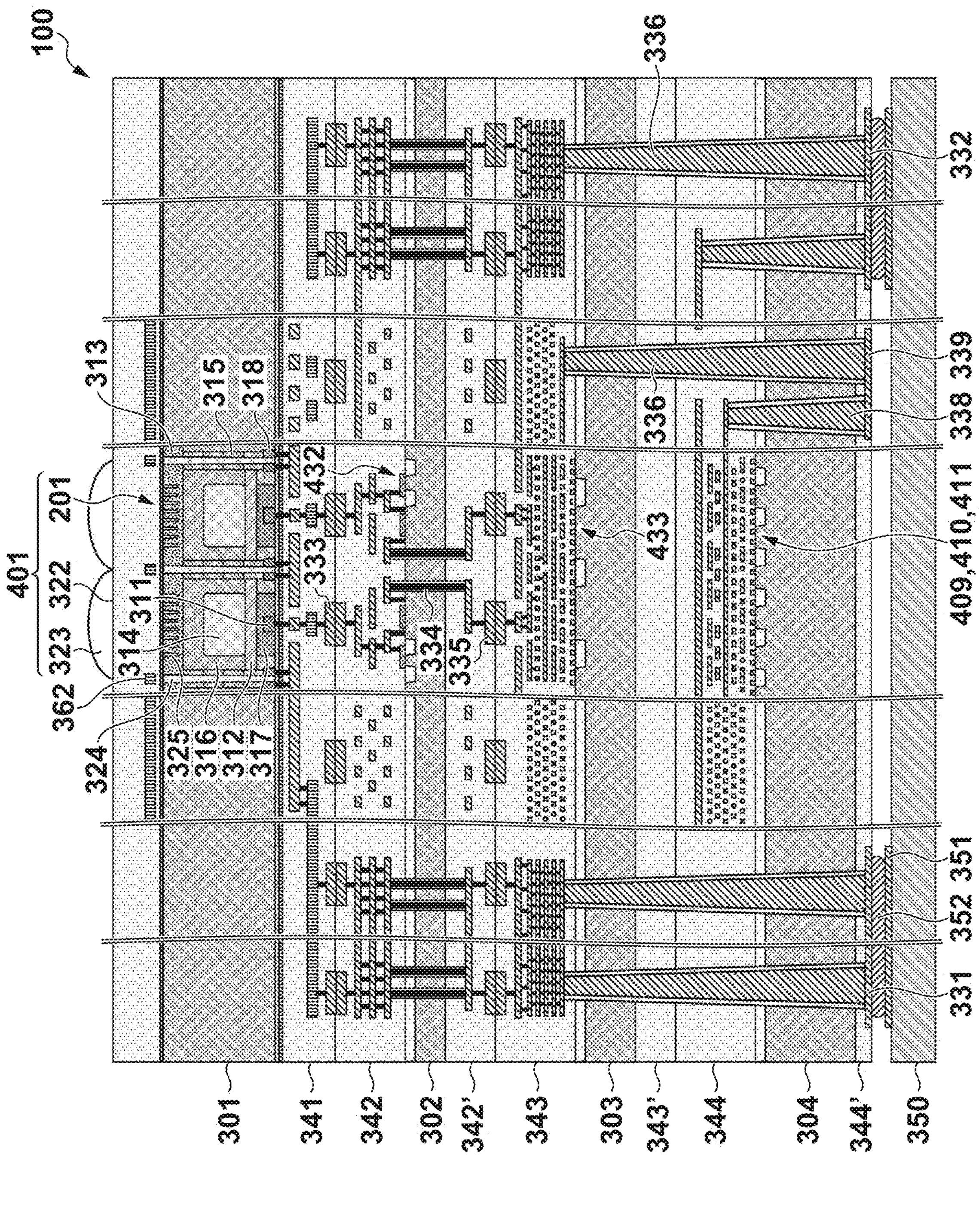

100
401
201
322
323
313
315
318
311
314
362
324
325
316
312
317
333
334
335
432
433
336
338
339
332
331
352
351
409,410,411
301
341
342
302
342'
343
303
343'
344
304
344'
350

FIG. 11

FIG. 12

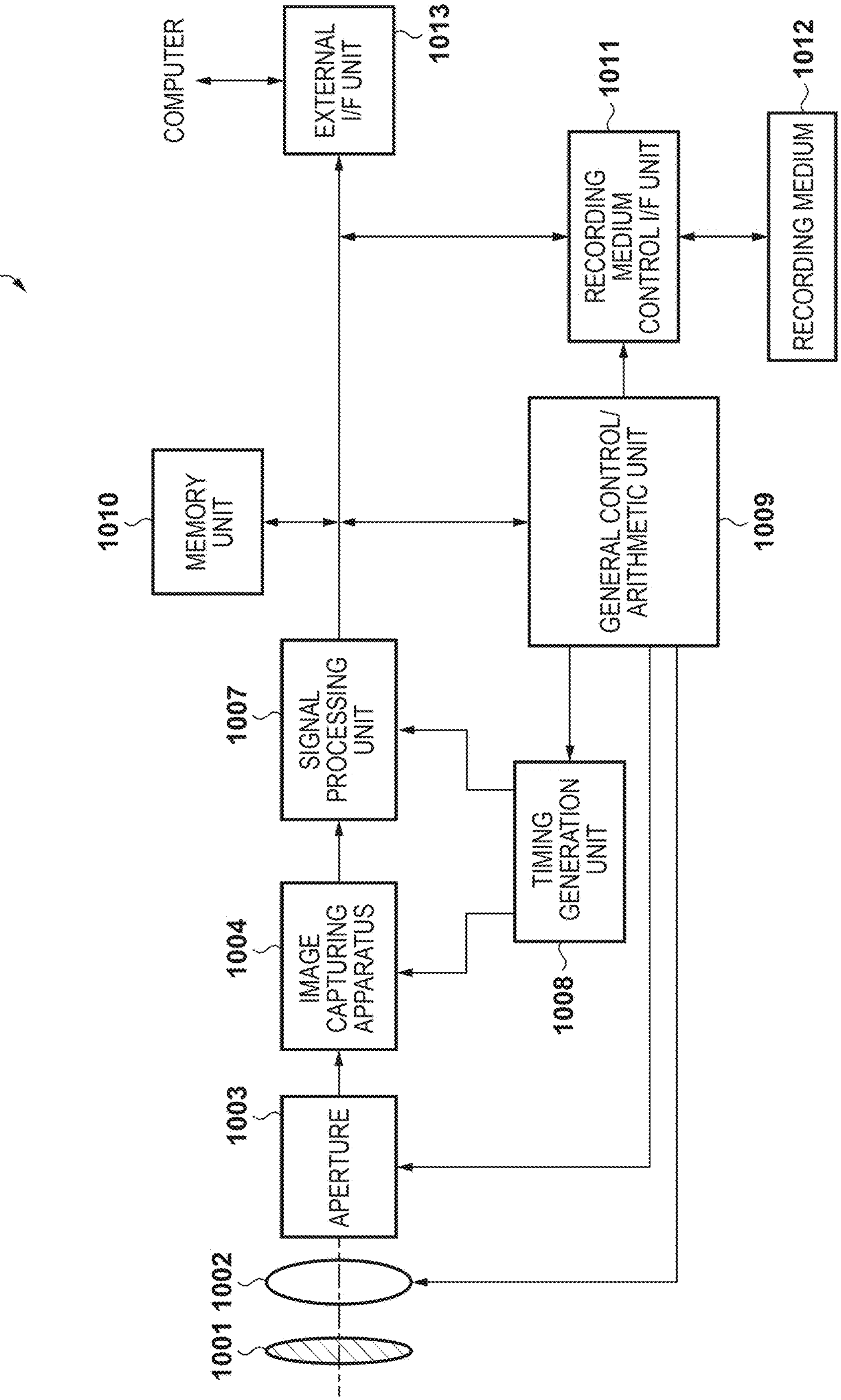

1000
COMPUTER
EXTERNAL I/F UNIT
1013
1011
RECORDING MEDIUM CONTROL I/F UNIT
1012
RECORDING MEDIUM
1010
MEMORY UNIT
GENERAL CONTROL/ ARITHMETIC UNIT
1009
1007
SIGNAL PROCESSING UNIT
TIMING GENERATION UNIT
1008
1004
IMAGE CAPTURING APPARATUS
1003
APERTURE
1001
1002

FIG. 13A

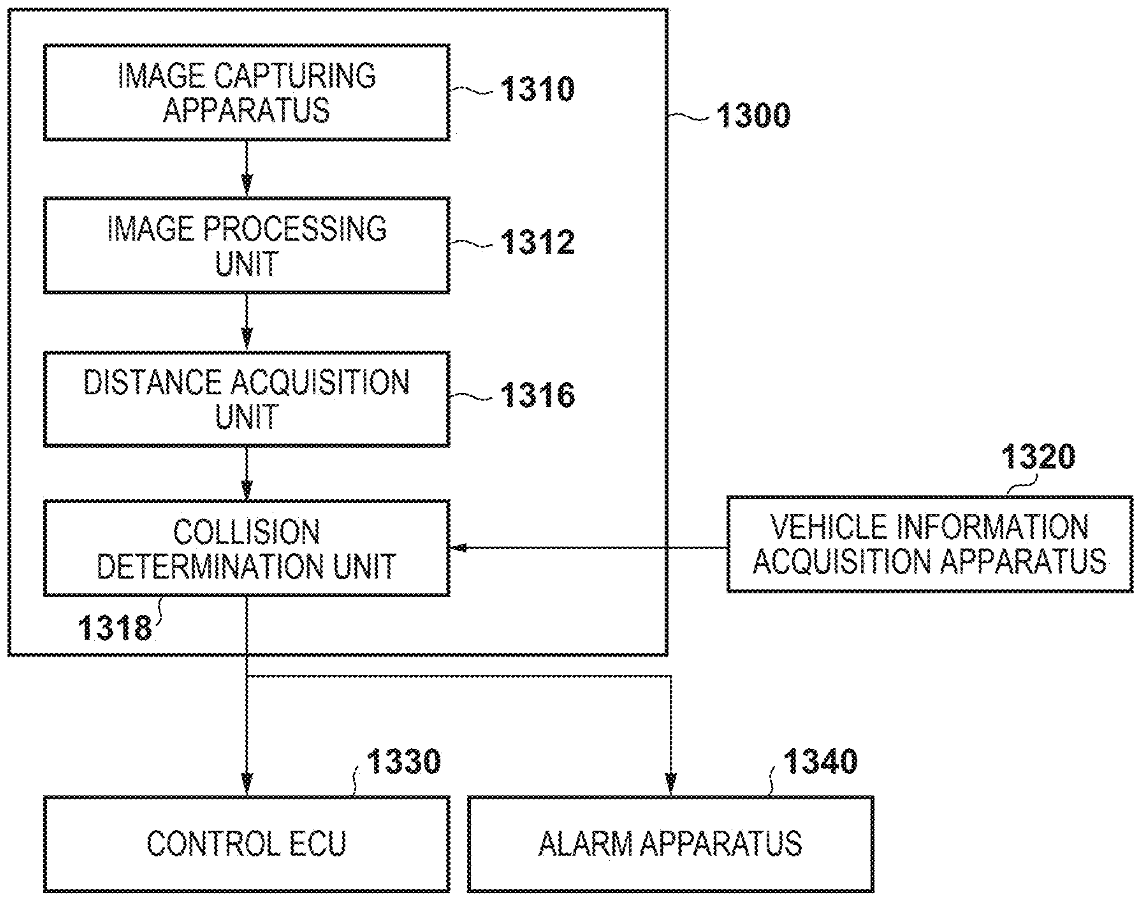
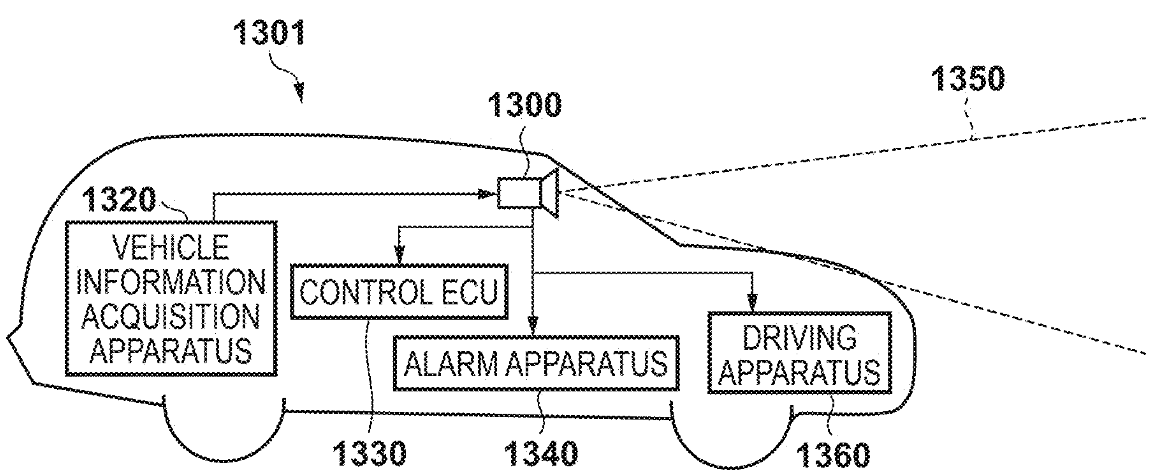


FIG. 13B



F I G. 14

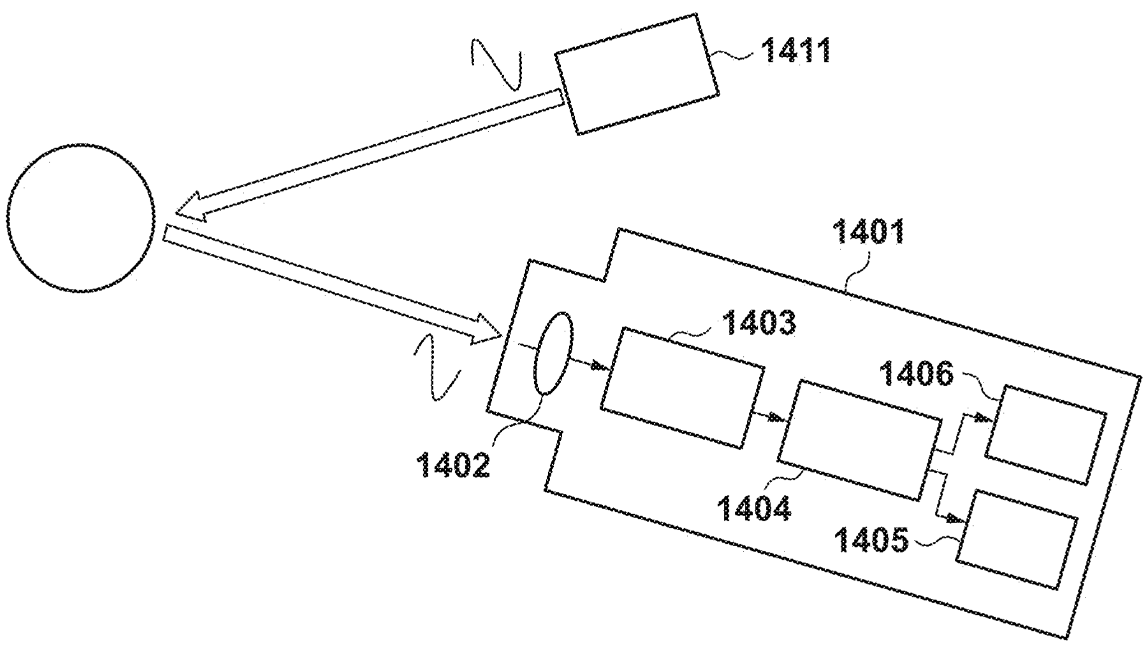
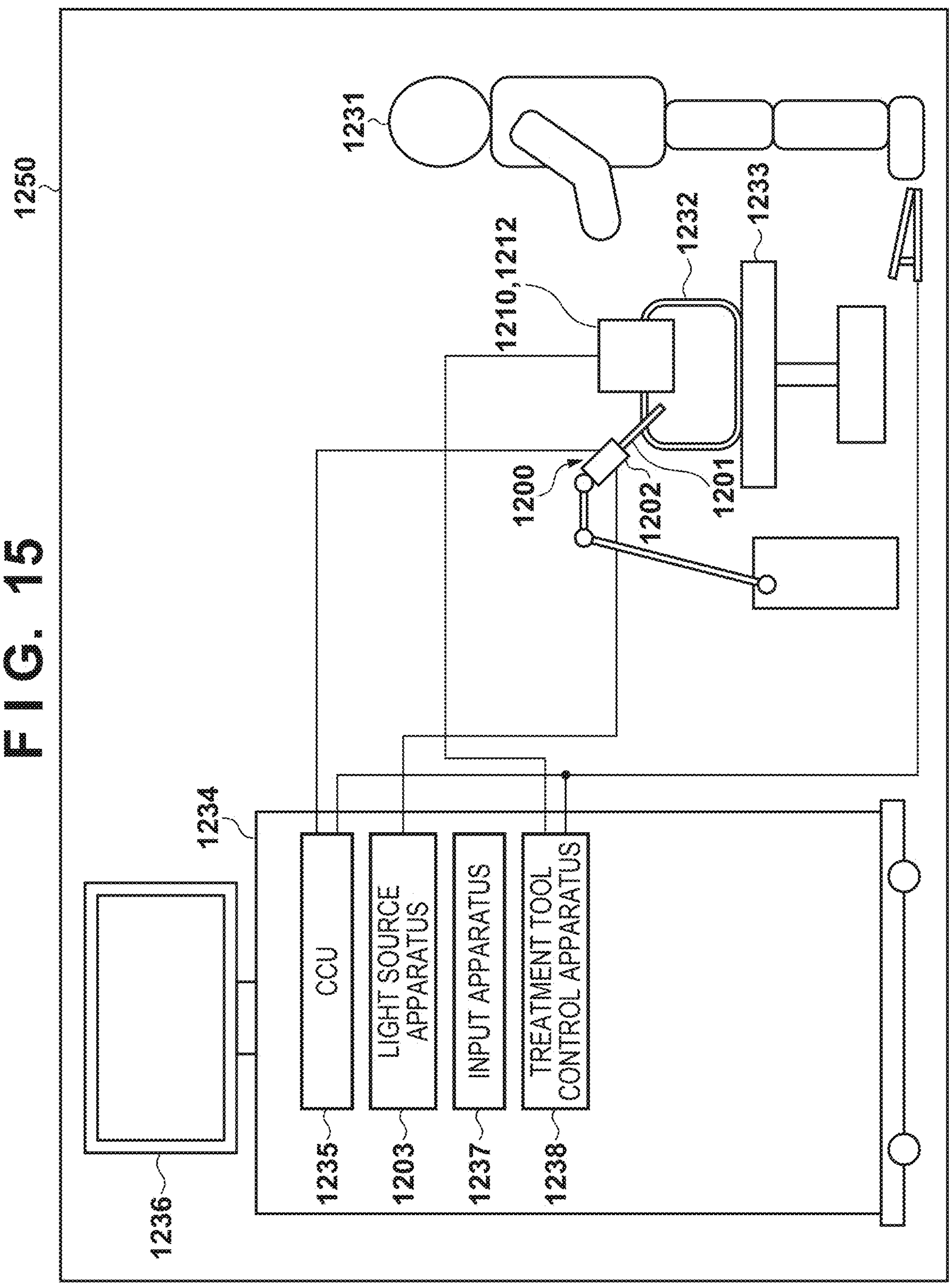
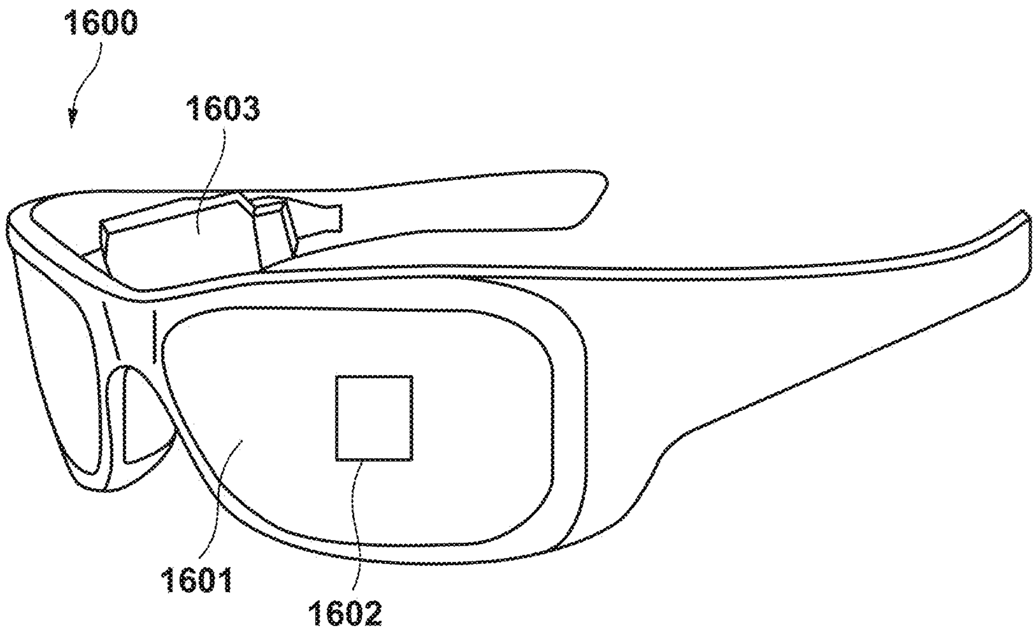


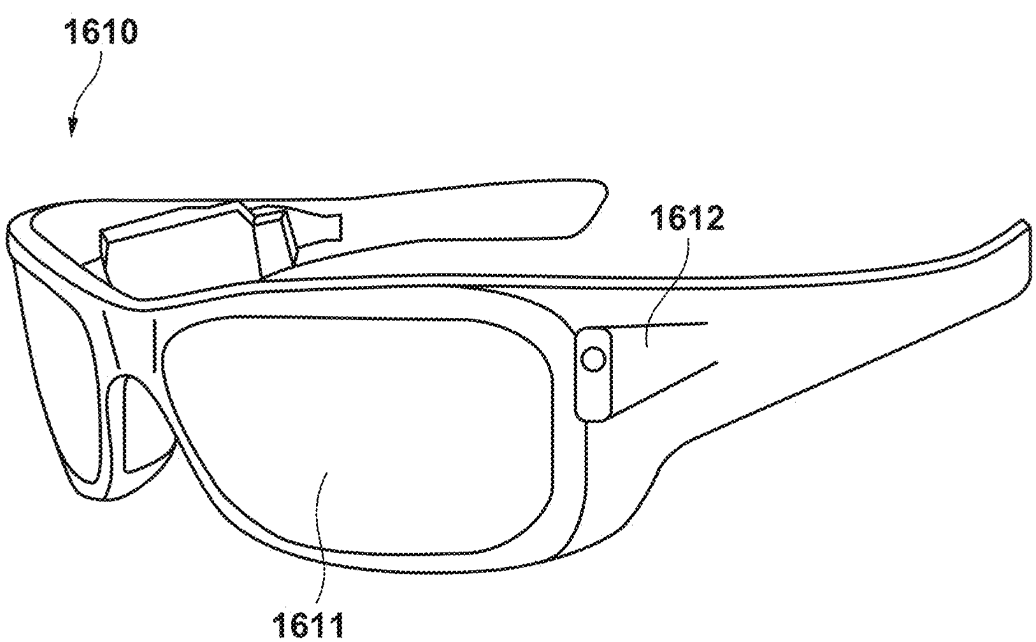
FIG. 15


1250
1231
1232
1233
1210, 1212
1200
1202
1201
1234
CCU
LIGHT SOURCE APPARATUS
INPUT APPARATUS
TREATMENT TOOL CONTROL APPARATUS
1235
1203
1237
1238
1236

F I G. 16A



F I G. 16B



PHOTOELECTRIC CONVERSION APPARATUS, PHOTOELECTRIC CONVERSION SYSTEM, MOVING BODY, AND EQUIPMENT

BACKGROUND

Technical Field

[0001] The present disclosure relates to a photoelectric conversion apparatus, a photoelectric conversion system, a moving body, and equipment.

Description of the Related Art

[0002] An avalanche photodiode (APD) capable of detecting light at the single photon level is known. Japanese Patent Laid-Open No. 2022-113123 discloses a photoelectric conversion apparatus in which a first substrate including a plurality of pixels with APDs, a second substrate including a plurality of pixel circuits provided in correspondence with the plurality of pixels, and a third substrate including a signal processing circuit that processes signals output from the plurality of pixel circuits are stacked.

[0003] A photoelectric conversion apparatus is demanded to increase the number of pixels, achieve integration, and improve functionality.

SUMMARY

[0004] According to some embodiments, a back-illuminated type photoelectric conversion apparatus in which a first semiconductor layer including an APD array in which a plurality of avalanche photodiodes (APDs) are arranged, a second semiconductor layer including a plurality of first pixel circuits respectively for the plurality of APDs, a third semiconductor layer including a plurality of second pixel circuits respectively for the plurality of first pixel circuits, and a fourth semiconductor layer including a processing circuit configured to process signals output from the plurality of second pixel circuits are stacked, wherein the second semiconductor layer is arranged between the first semiconductor layer and the fourth semiconductor layer, the third semiconductor layer is arranged between the second semiconductor layer and the fourth semiconductor layer, and a through via extending through the third semiconductor layer is arranged in a region not overlapping the APD array, is provided.

[0005] According to some embodiments of the present disclosure, it is possible to provide a technique advantageous in improving performance of a photoelectric conversion apparatus.

[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a view showing an example of the configuration of a photoelectric conversion apparatus according to an embodiment.

[0008] FIGS. **2**A to **2**D are views each showing an example of the configuration of each semiconductor layer of the photoelectric conversion apparatus shown in FIG. **1**.

[0009] FIG. **3** is a circuit diagram showing an example of the configuration of the pixel of the photoelectric conversion apparatus shown in FIG. **1**.

[0010] FIGS. **4**A and **4**B show a circuit diagram and timing charts for explaining an example of the operation of the pixel of the photoelectric conversion apparatus shown in FIG. **1**.

[0011] FIGS. **5**A to **5**I are views for explaining selection of the stacked structure of the photoelectric conversion apparatus shown in FIG. **1**.

[0012] FIG. **6** is a sectional view showing an example of the configuration of the photoelectric conversion apparatus shown in FIG. **1**.

[0013] FIG. **7** is a sectional view showing an example of the manufacturing step of the photoelectric conversion apparatus shown in FIG. **6**.

[0014] FIG. **8** is a sectional view showing an example of the configuration of the photoelectric conversion apparatus shown in FIG. **1**.

[0015] FIG. **9** is a sectional view showing an example of the manufacturing step of the photoelectric conversion apparatus shown in FIG. **8**.

[0016] FIG. **10** is a sectional view showing an example of the configuration of the photoelectric conversion apparatus shown in FIG. **1**.

[0017] FIG. **11** is a sectional view showing an example of the configuration of the photoelectric conversion apparatus shown in FIG. **1**.

[0018] FIG. **12** is a block diagram showing a photoelectric conversion system using the photoelectric conversion apparatus according to the embodiment.

[0019] FIGS. **13**A and **13**B are a block diagram and a view showing a photoelectric conversion system using the photoelectric conversion apparatus according to the embodiment.

[0020] FIG. **14** is a view showing a photoelectric conversion system using the photoelectric conversion apparatus according to the embodiment.

[0021] FIG. **15** is a view showing a photoelectric conversion system using the photoelectric conversion apparatus according to the embodiment.

[0022] FIGS. **16**A and **16**B are views each showing a photoelectric conversion system using the photoelectric conversion apparatus according to the embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0024] A photoelectric conversion apparatus according to an embodiment of the present disclosure will be described first with reference to FIGS. **1** to **11**. FIG. **1** is a view showing an example of the configuration of a photoelectric conversion apparatus **100** according to this embodiment.

[0025] The photoelectric conversion apparatus **100** is formed by stacking four semiconductor layers **301** to **304** (to be also referred to as substrates hereinafter) and electrically connecting them. The semiconductor layer **301** includes an APD array **401** in which a plurality of avalanche photodiodes (APDs) **201** (shown in FIG. **2**A) are arranged. The semiconductor layer **302** includes a pixel circuit array **402** in which a plurality of pixel circuits respectively for the plurality of APDs **201** are arranged. The semiconductor layer **303** includes a pixel circuit array **403** in which a plurality of pixel circuits respectively for the plurality of pixel circuits arranged in the semiconductor layer **302** are arranged. In addition, a signal transfer unit **406** for transferring signals output from the plurality of pixel circuits arranged in the semiconductor layer **303** to the semiconductor layer **304** is arranged in the semiconductor layer **303**. The semiconductor layer **304** includes a processing circuit **411** for processing signals output from the plurality of pixel circuits arranged in the semiconductor layer **303**. In addition, memories **409** and **410** that store signals output from the plurality of pixel circuits arranged in the semiconductor layer **303** and the like are arranged in the semiconductor layer **304**. As shown in FIG. **1**, the semiconductor layer **302** is arranged between the semiconductor layers **301** and **304** and the semiconductor layer **303** is arranged between the semiconductor layers **302** and **304**.

[0026] FIGS. **2**A to **2**D are views each showing an example of the arrangement of components in each of the semiconductor layers **301** to **304**. As shown in FIG. **2**A, the semiconductor layer **301** includes the APD array **401** in which the plurality of APDs **201** are arranged to form a plurality of rows and a plurality of columns. The APD **201** is typically arranged to generate an image but need not always generate an image when it is used in a Time of Flight (ToF) distance measurement device. That is, the APD **201** may be configured to measure the time when light arrives and the amount of light. Furthermore, as shown in FIG. **2**A, a connection portion **333** for connecting the APD **201** arranged in the semiconductor layer **301** and a pixel circuit **432** arranged in the semiconductor layer **302** is arranged between the semiconductor layers **301** and **302** to overlap the APD **201**. The arrangement of the connection portion **333** will be described later.

[0027] As shown in FIG. **2**B, the semiconductor layer **302** includes the pixel circuit array **402** in which the plurality of pixel circuits **432** are arranged to form a plurality of rows and a plurality of columns. With respect to the plurality of pixel circuits **432**, one pixel circuit **432** may be arranged in correspondence with each of the plurality of APDs **201**. This specification assumes that one APD **201** and one pixel circuit **432** have a correspondence relationship but the present invention is not limited to this. One pixel circuit **432** may be arranged in correspondence with one group constituted by a predetermined number of APDs **201** among the plurality of APDs **201**. Details of the pixel circuit **432** will be described later. Furthermore, as shown in FIG. **2**B, a connection portion **335** for connecting the pixel circuit **432** arranged in the semiconductor layer **302** and a pixel circuit **433** arranged in the semiconductor layer **303** is arranged between the semiconductor layers **302** and **303** to overlap the pixel circuit **432**. The arrangement of the connection portion **335** will be described later.

[0028] As shown in FIG. **2**B, a temperature measurement unit **413** may be provided in the semiconductor layer **302**. The temperature measurement unit **413** can be a sensor for measuring the temperature. The characteristic of the APD **201** may change along with a change of the temperature. Therefore, the temperature may be measured in the semiconductor layer **302** close to the semiconductor layer **301** in which the APD **201** is arranged, and may be applied to driving control of the APD **201**, control of a voltage to be applied to the APD **201**, correction of a signal output from the APD **201**, and the like. The temperature measurement unit **413** may be arranged in the semiconductor layer **301**.

[0029] As shown in FIG. **2**C, the semiconductor layer **303** includes the pixel circuit array **403** in which the plurality of pixel circuits **433** are arranged to form a plurality of rows and a plurality of columns. With respect to the plurality of pixel circuits **433**, one pixel circuit **433** may be arranged in correspondence with each of the plurality of pixel circuits **432**. Furthermore, for example, one pixel circuit **433** may be arranged in correspondence with one group constituted by a predetermined number of pixel circuits **432** among the plurality of pixel circuits **432**. As described above, the signal transfer unit **406** for transferring the signals output from the plurality of pixel circuits **433** arranged in the semiconductor layer **303** to the semiconductor layer **304** is arranged in the semiconductor layer **303**. As shown in FIG. **2**C, connection portions **337** for connecting the signal transfer unit **406** arranged in the semiconductor layer **303** and a signal reception unit **407** arranged in the semiconductor layer **304** are arranged between the semiconductor layers **303** and **304** to overlap the signal transfer unit **406**. The arrangement of the connection portions **337** will be described later.

[0030] A scanning circuit **404** and a readout circuit **405** are further arranged in the semiconductor layer **303**. In the configurations shown in FIG. **2**A to **2**C, one APD **201**, one pixel circuit **432**, and one pixel circuit **433** are combined to form one so-called "pixel". The scanning circuit **404** and the readout circuit **405** operate each pixel in cooperation with each other, thereby operating to read out a signal from the pixel. The scanning circuit **404** and the readout circuit **405** can be controlled by, for example, a control unit **408** arranged in the semiconductor layer **304** shown in FIG. **2**D. In FIGS. **2**C and **2**D, control of each pixel formed by the APD **201**, the pixel circuit **432**, and the pixel circuit **433** and the flow of the signal from each pixel are indicated by arrows.

[0031] As shown in FIG. **2**D, the semiconductor layer **304** includes the processing circuit **411** for processing the signals output from the plurality of pixel circuits **433** arranged in the semiconductor layer **303**. It can be said that the processing circuit **411** processes the signal output from the pixel formed by the APD **201**, the pixel circuit **432**, and the pixel circuit **433**. Furthermore, the memories **409** and **410** that store the signals and the like are arranged in the semiconductor layer **304**. In the configuration shown in FIG. **2**D, the two memories **409** and **410** are arranged in the semiconductor layer **304**. However, only one memory or three or more memories may be arranged.

[0032] In addition, the signal reception unit **407**, the control unit **408**, and an output unit **412** are arranged in the semiconductor layer **304**. The signal reception unit **407** receives the signal output from the signal transfer unit **406** arranged in the semiconductor layer **303**. In a planar view, the signal transfer unit **406** and the signal reception unit **407** may be arranged along the long side of the APD array **401**, as shown in FIGS. **2**A, **2**C, and **2**D. This makes it possible to efficiently arrange more through vias and the more connection portions **337** for connecting the signal transfer unit **406** and the signal reception unit **407**, as compared to a case where the signal transfer unit **406** and the signal reception unit **407** are arranged along the short side of the APD array **401**. As a result, it is possible to increase the number of parallel transmissions of the signals output from the pixels each formed by the APD **201**, the pixel circuit **432**, and the pixel circuit **433**, thereby speeding up data transmission. The control unit **408** controls the respective components arranged in the photoelectric conversion apparatus **100**. Thus, the signal reception unit **407** can function as an output unit for transferring a control signal output from the control unit **408** to the semiconductor layers **301** to **303**. The output unit **412** is arranged to output a signal generated by the photoelectric conversion apparatus **100** to the outside of the photoelectric conversion apparatus **100**. In addition, the output unit **412** may function as an input unit that receives, from the outside of the photoelectric conversion apparatus **100**, a signal for controlling the photoelectric conversion apparatus **100**. That is, the output unit **412** can function as an interface between the photoelectric conversion apparatus **100** and an apparatus arranged outside the photoelectric conversion apparatus **100**.

[0033] FIG. **3** is a circuit diagram including an equivalent circuit that focuses on four pixels (each pixel is formed by a combination of the APD **201**, the pixel circuit **432**, and the pixel circuit **433**). As shown in FIG. **3**, in each pixel, the APD **201** is arranged in the semiconductor layer **301**, the pixel circuit **432** is arranged in the semiconductor layer **302**, and the pixel circuit **433** is arranged in the semiconductor layer **303**. The pixel circuit **433** is connected to the readout circuit **405** arranged in the semiconductor layer **303**. The readout circuit **405** is connected to the memories **409** and **410** and the processing circuit **411**, all of which are arranged in the semiconductor layer **304**.

[0034] The APD **201** generates a charge pair corresponding to incident light by photoelectric conversion. A potential VL is supplied to the anode of the APD **201**. A potential VH higher than the potential VL supplied to the anode is supplied to the cathode of the APD **201**. A reverse bias voltage is supplied to the anode and the cathode so that the APD **201** performs an avalanche breakdown operation. In a state in which such a reverse bias voltage is supplied, charges generated by incident light cause avalanche breakdown, generating an avalanche current.

[0035] In a case where the reverse bias voltage is supplied to the APD **201**, there are a Geiger mode in which the APD **201** is operated by a potential difference (voltage) between the anode and the cathode larger than a breakdown voltage, and a linear mode in which the APD **201** is operated by a potential difference between the anode and the cathode around the breakdown voltage or equal to or lower than the breakdown voltage. An APD operated in the Geiger mode is called a Single Photon Avalanche Diode (SPAD). For example, the potential VL is-30 V, and the potential VH is 1 V. The APD **201** may be operated in the linear mode or the Geiger mode.

[0036] A quench circuit **202** and a level shift circuit **220** are arranged as the pixel circuit **432** in the semiconductor layer **302**. The quench circuit **202** is connected between the APD **201** and a power supply that supplies the potential VH. The quench circuit **202** functions as a load circuit at the time of signal multiplication by avalanche breakdown, and operates to suppress a voltage supplied to the APD **201** and suppress avalanche breakdown (quench operation). The quench circuit **202** also operates to return the voltage supplied to the APD **201** to a voltage (VH−VL) by supplying a current by an amount corresponding to a voltage drop caused by the quench operation (recharge operation).

[0037] The level shift circuit **220** shapes a potential change of the cathode of the APD **201** obtained at the time of photon detection to a signal that can be detected by the pixel circuit **433** of the succeeding stage. For example, the level shift circuit **220** outputs a pulse signal corresponding to the potential change of the APD **201**. As the level shift circuit **220**, for example, an inverter circuit may be used. However, the present invention is not limited to this, and another circuit may be used as the level shift circuit **220** as long as the circuit can shape the potential change of the cathode of the APD **201** obtained at the time of photon detection to a signal that can be detected by the pixel circuit **433**.

[0038] A counter circuit or the like can be arranged as the pixel circuit **433** in the semiconductor layer **303**. The counter circuit counts pulse signals output from the level shift circuit **220** and holds the count value. Furthermore, a selection circuit that switches electrical connection/disconnection between the counter circuit and a signal line **263** for outputting a signal to the readout circuit **405** may be arranged as the pixel circuit **433**. The selection circuit can include, for example, a buffer circuit for outputting a signal. In addition, a reset circuit for resetting the count value counted by the counter circuit or the like may be arranged as the pixel circuit **433** in the semiconductor layer **303**.

[0039] In this embodiment, the counter circuit is arranged as the pixel circuit **433**. However, the present invention is not limited to this. For example, a Time-to-Digital Converter (TDC) and a memory may be arranged as the pixel circuit **433** instead of the counter circuit so that the photoelectric conversion apparatus **100** obtains a pulse detection timing. In this case, the generation timing of a pulse signal output from the level shift circuit **220** is converted into a digital signal by the TDC. The TDC is supplied with a reference signal for measurement of the timing of the pulse signal. By using the reference signal as a reference, the TDC obtains, as a digital signal, a signal when the input timing of a signal indicating the potential change of the APD **201** via the level shift circuit **220** is regarded as a relative time.

[0040] FIGS. **4**A and **4**B show a circuit diagram and timing charts schematically showing the relationship between the operation of the APD **201** and the output signal. FIG. **4**A is a circuit diagram showing an excerpt of the APD **201**, the quench circuit **202**, and the level shift circuit **220** shown in FIG. **3**. The input side of the level shift circuit **220** is a node A and the output side of the level shift circuit **220** is a node B. FIG. **4**B shows waveform changes at the node A and the node B.

[0041] From time t**0** to time t**1**, a potential difference (voltage) of the potential VH−the potential VL is applied to the APD **201**. When a photon enters the APD **201** at time t**1**, avalanche breakdown occurs in the APD **201**, an avalanche breakdown current flows into the quench circuit **202**, and the potential of the node A drops. When the voltage drop amount further increases and the potential difference applied to the APD **201** decreases, the avalanche breakdown of the APD **201** stops as shown at time t**2**, and the potential level of the node A does not drop any more from a predetermined value. After that, in a period from time t**2** to time t**3**, a current compensating for the voltage drop from the potential VL flows to the node A. At time t**3**, the node A is settled at the original potential level. At this time, a portion at which the output waveform exceeds a given threshold at the node A is waveform-shaped by the level shift circuit **220** and output as a signal to the node B.

[0042] The photoelectric conversion apparatus **100** is demanded to miniaturize the pixel, increase the number of pixels, achieve integration, and improve functionality. The photoelectric conversion apparatus **100** according to this embodiment stacks the four semiconductor layers **301** to **304**. Among the APD **201**, the pixel circuit **432**, and the pixel circuit **433** forming one pixel, the APD **201** is arranged in the semiconductor layer **301**, the pixel circuit **432** is arranged in the semiconductor layer **302**, and the pixel circuit **433** is arranged in the semiconductor layer **303**. This can decrease the number of components of one pixel and reduce its size in the semiconductor layers **301** to **303**. As a result, miniaturization of the pixel is possible. In addition, since each pixel can be made small, a larger number of pixels can be obtained.

[0043] Furthermore, no pixel circuit forming the pixel is arranged in the semiconductor layer **304**. Therefore, it is possible to provide a large area to the memories **409** and **410** and the processing circuit **411** for processing the signal output from the pixel. That is, it is possible to execute complicated processing of performing more calculation operations. That is, the functionality of the photoelectric conversion apparatus **100** can be improved.

[0044] The signals from the pixels distributed and arranged in the semiconductor layers **301** to **303** are stored as, for example, image data in the memories **409** and **410**. The processing circuit **411** may execute various signal processes for the image data read out from the memory **409** or **410**. For example, if the image data to be processed is color image data, the processing circuit **411** may convert the format into YUV image data or RGB image data, and output it via the output unit **412**. Furthermore, for example, the processing circuit **411** may execute processing such as noise removal and white balance adjustment for the image data read out from the memory **409** or **410**, as needed.

[0045] For example, image data of even-numbered frames are stored in the memory **409** and image data of odd-numbered frames are stored in the memory **410**. While the storage operation of the image data in the memory **409** (or the memory **410**) is performed, the processing circuit **411** may read out the image data from the memory **410** (or the memory **409**) and perform processing. That is, the memories **409** and **410** may be frame memories. Furthermore, for example, the processing circuit **411** may process the image data written in the memory **409**, and the processed image data may be stored in the memory **410**. The processing circuit **411** may execute calculation processing using the image data based on the signals output from the pixels and the processed image data based on the signals obtained by processing the signals output from the pixels. Therefore, in consideration of the flow of the signals, the processing circuit **411** may be arranged between the memories **409** and **410**, as shown in FIG. **2**D.

[0046] If the TDC is arranged as the pixel circuit **433** and the photoelectric conversion apparatus **100** is used as a distance measurement device, the processing circuit **411** can also function as a distance measurement processing unit. For example, the processing circuit **411** may create a histogram based on information obtained from the TDC, and perform distance measurement calculation. The histogram is represented such that the abscissa represents a class (bin) related to the time and the ordinate represents a frequency at each class. The frequency indicates the number of times light is received during a predetermined light-receiving time. A count based on reflected light and ambient light is mixed in the histogram. Thus, by setting a predetermined threshold, the count of the reflected light component and the count of the ambient light component are separated. The distance between the distance measurement device and a measurement target object is calculated from a light arrival time corresponding to the reflected light component.

[0047] Furthermore, the processing circuit **411** may execute a program stored in the memory **409** or **410**, thereby executing various processes using a learned model created by machine learning. For example, the learned model is created by machine learning using a deep neural network (DNN). The learned model is also called a neural network calculation model.

[0048] The learned model may be designed based on a parameter generated by inputting the signal output from the pixel and learning data linked with a label corresponding to the signal output from the pixel to a predetermined machine learning model. The predetermined machine learning model may be a learning model using a multilayer neural network. This learned model is also called a multilayer neural network model.

[0049] For example, the processing circuit **411** executes calculation processing based on the learned model stored in the memory **409** or **410**. The calculation result is output to the memory **409** or **410** or the like. The calculation result includes image data obtained by executing the calculation processing using the learned model and various kinds of information (metadata) obtained from the image data.

[0050] The image data to be processed by the processing circuit **411** may be image data normally read out from the pixel, or image data whose data size is reduced by thinning out some data from the image data. The image data to be processed by the processing circuit **411** may be three-dimensional distance image data. With the three-dimensional distance image data, it is possible to accurately recognize an object and acquire accurate position information of the object since the information amount of the three-dimensional distance image data is larger than that of two-dimensional image data.

[0051] As described above, in the semiconductor layer **304**, an area where the processing circuit **411** is arranged can be made large. Therefore, it is possible to mount the learned model for performing calculation with a heavy load, and execute processing of three-dimensional distance information data. This improves the functionality and performance of the photoelectric conversion apparatus **100**.

[0052] Consider the stacked structure of the photoelectric conversion apparatus **100** according to this embodiment. FIGS. **5**A to **5**I are views for explaining the arrangement of surfaces on which transistors and the like forming the above-described pixel circuits **432** and **433** and processing circuit **411** are arranged in the semiconductor layers **301** to **304**. FIG. **5**A shows the semiconductor layers **301** to **304**, semiconductor device forming regions **340** formed in the semiconductor layers **301** to **304**, and wiring layers **341** to **344** such as wiring patterns connected to the semiconductor devices arranged in the forming regions **340**. In this embodiment, the photoelectric conversion apparatus **100** is a back-illuminated type photoelectric conversion apparatus, as shown in FIGS. **5**B to **5**I. Therefore, the APD **201** is connected to the wiring pattern arranged in the wiring layer **341** arranged on the opposite side of an optical layer **345** including a microlens on the light incident side of the photoelectric conversion apparatus **100**.

[0053] Next, consider the directions of the semiconductor device forming regions **340** arranged in the semiconductor layers **302** to **304**. In the following description, the light incident side of the photoelectric conversion apparatus **100** on which the optical layer **345** is arranged is set as the "upper side". Therefore, for example, the semiconductor layer **302** is arranged on the "upper side" of the semiconductor layer **303**, and is arranged on the "lower side" of the semiconductor layer **301**. In the configuration shown in FIG. **5**B, the semiconductor device forming region **340** in each of the semiconductor layers **302** to **304** faces upward. In the configuration shown in FIG. **5**C, the semiconductor device forming region **340** in the semiconductor layer **302** faces downward, and the semiconductor device forming regions **340** in the semiconductor layers **303** and **304** face upward. In the configuration shown in FIG. **5**D, the semiconductor device forming regions **340** in the semiconductor layers **302** and **304** face upward, and the semiconductor device forming region **340** in the semiconductor layer **303** faces downward. In the configuration shown in FIG. **5**E, the semiconductor device forming regions **340** in the semiconductor layers **302** and **303** face downward, and the semiconductor device forming region **340** in the semiconductor layer **304** faces upward. In the configuration shown in FIG. **5**F, the semiconductor device forming regions **340** in the semiconductor layers **302** and **303** face upward, and the semiconductor device forming region **340** in the semiconductor layer **304** faces downward. In the configuration shown in FIG. **5**G, the semiconductor device forming regions **340** in the semiconductor layers **302** and **304** face downward, and the semiconductor device forming region **340** in the semiconductor layer **303** faces upward. In the configuration shown in FIG. **5**H, the semiconductor device forming region **340** in the semiconductor layer **302** faces upward, and the semiconductor device forming regions **340** in the semiconductor layers **303** and **304** face downward. In the configuration shown in FIG. **5**I, the semiconductor device forming region **340** in each of the semiconductor layers **302** to **304** faces downward.

[0054] As described above, the APD **201** arranged in the semiconductor layer **301** and the pixel circuit **432** arranged in the semiconductor layer **302** are in one-to-one correspondence with each other. The pixel circuit **432** arranged in the semiconductor layer **302** and the pixel circuit **433** arranged in the semiconductor layer **303** are in one-to-one correspondence with each other. The pixel circuit **432** is arranged in the semiconductor device forming region **340** in the semiconductor layer **302**. Similarly, the pixel circuit **433** is arranged in the semiconductor device forming region **340** in the semiconductor layer **303**. Therefore, in the pixel circuit array **402** of the semiconductor layer **302**, through vias the number of which is equal to the number of pixels each formed by the APD **201** and the pixel circuits **432** and **433** are arranged regardless of the direction in which the semiconductor device forming region **340** in the semiconductor layer **302** is arranged.

[0055] Next, consider the direction of the semiconductor device forming region **340** in which the pixel circuit **433** and the like of the semiconductor layer **303** are arranged. As in the configuration shown in each of FIGS. **5**D, **5**E, **5**H, and **5**I, if the semiconductor device forming region **340** in the semiconductor layer **301** faces downward, through vias the number of which is equal to the number of pixels are necessary for the pixel circuit array **403** of the semiconductor layer **303**. Since it is difficult to arrange semiconductor devices in a region where the through vias are arranged, this may be inappropriate for integration of the pixel circuits **433**. On the other hand, as in the configuration shown in each of FIGS. **5**B, **5**C, **5**F, and **5**G, if the semiconductor device (pixel circuit **433**) forming region **340** in the semiconductor layer **301** faces upward, it is unnecessary to form through vias in the pixel circuit array **403** of the semiconductor layer **303**. Therefore, a case where the semiconductor device forming region **340** in the semiconductor layer **303** faces upward is more suitable for integration than in a case where the semiconductor device forming region **340** in the semiconductor layer **303** faces downward. That is, the semiconductor devices forming the pixel circuit **433** and the like may be arranged on the surface of the semiconductor layer **303** on the side of the semiconductor layer **302**.

[0056] Following the direction of the semiconductor device forming region **340** in the semiconductor layer **303**, consider the direction of the semiconductor device forming region **340** in which the processing circuit **411** and the memories **409** and **410** of the semiconductor layer **304** are arranged. As in the configuration shown in each of FIGS. **5**F to **5**I, if the semiconductor device forming region **340** in the semiconductor layer **301** faces downward, the parasitic capacitance of the wiring pattern for transferring the signal from the semiconductor layer **303** to the semiconductor layer **304** increases, thereby making it difficult to increase the speed. On the other hand, as in the configuration shown in each of FIGS. **5**B to **5**E, if the semiconductor device forming region **340** in the semiconductor layer **301** faces upward, the wiring pattern for connecting the semiconductor layers **303** and **304** becomes shorter than in a case where the semiconductor device forming region **340** in the semiconductor layer **301** faces downward. As a result, the parasitic capacitance of the wiring pattern for transferring the signal from the semiconductor layer **303** to the semiconductor layer **304** decreases, thereby making it easy to increase the speed. That is, the semiconductor devices forming the processing circuit **411** and the like may be arranged on the surface of the semiconductor layer **304** on the side of the semiconductor layer **303**.

[0057] In consideration of the above description, as shown in FIGS. **5**B and **5**C, a case where the semiconductor device forming region **340** in each of the semiconductor layers **303** and **304** faces upward is more suitable for improving the performance of the photoelectric conversion apparatus **100**.

[0058] FIG. **6** is a sectional view showing the more detailed configuration of the photoelectric conversion apparatus **100** shown in FIG. **5**B. The APD **201** is arranged in the semiconductor layer **301**. The APD **201** includes semiconductor regions **311**, **314**, **316**, and **317** of the same conductivity type. Furthermore, the APD **201** includes semiconductor regions **312**, **313**, **315**, and **318** of a conductivity type opposite to that of the semiconductor regions **311**, **314**, **316**, and **317**. For example, the semiconductor regions **311**, **314**, **316**, and **317** can be n-type semiconductor regions and the semiconductor regions **312**, **313**, **315**, and **318** can be p-type semiconductor regions. An example of a semiconductor material used for the semiconductor layers **301** to **304** is silicon. Therefore, each of the semiconductor regions **311** to

318 can be a region where an impurity corresponding to the conductivity type is implanted into silicon. For example, each of the semiconductor regions 311 to 315, 317, and 318 may be formed in the n-type semiconductor layer 301 (semiconductor region 316) using an ion implantation method or the like.

[0059] The semiconductor region 311 is a region having a higher n-type impurity concentration than the semiconductor regions 314 and 317. A p-n junction portion is formed between the p-type semiconductor region 312 and the n-type semiconductor region 311. By setting the impurity concentration of the semiconductor region 312 lower than the impurity concentration of the semiconductor region 311, a reverse bias is applied and the whole region of the semiconductor region 312 overlapping the center of the semiconductor region 311 becomes a depletion layer region. In this case, the potential difference between the semiconductor regions 311 and 312 is larger than the potential difference between the semiconductor regions 312 and 314. Furthermore, the depletion layer region extends to a partial region of the semiconductor region 311, and a strong electric field is induced in the depletion layer region. This strong electric field induces avalanche breakdown in the depletion layer region extending to the partial region of the semiconductor region 311, and a current based on the multiplied charges is output as a signal charge. When the light having entered the APD 201 is photoelectrically converted and avalanche breakdown is induced in the depletion layer region (avalanche breakdown region), the generated n-type charges are collected in the semiconductor region 311.

[0060] The semiconductor region 311 is connected to the pixel circuit 432 arranged in the semiconductor layer 302 via the wiring patterns arranged in the wiring layers 341 and 342 and the connection portions 333. The semiconductor region 312 is connected to a pad electrode 331 via the semiconductor regions 315 and 318 and the wiring pattern arranged in the wiring layer 341. By setting the impurity concentration of the semiconductor region 318 higher than the impurity concentration of the semiconductor region 315, the contact resistance between the semiconductor region 318 and the wiring pattern is reduced. As shown in FIG. 6, the pad electrode 331 is exposed via an opening formed in the semiconductor layer 301 between the semiconductor layers 301 and 302. The pad electrode 331 is connected to the semiconductor region 312 to function as the anode of the APD 201. The pad electrode 331 may be shared by the plurality of APDs 201.

[0061] The APDs 201 are isolated by a trench structure 324 provided in the semiconductor layer 301. A scattering diffraction structure 325 may be provided in the APD 201. The scattering diffraction structure 325 can be formed by forming trenches in the surface of the semiconductor layer 301 on the light incident side and embedding an appropriate material in the trenches.

[0062] A microlens 323 (the above-described optical layer 345) may be arranged on the semiconductor layer 301 via an insulating layer 322. A light shielding portion 326 for reducing a crosstalk between the adjacent APDs 201 may be arranged in the insulating layer 322. In addition, a color filter and the like may be arranged on the semiconductor layer 301.

[0063] The pixel circuit 432 arranged in the semiconductor layer 302 and the pixel circuit 433 arranged in the semiconductor layer 303 are connected via through vias 334 provided in the semiconductor layer 302, the wiring patterns provided in a wiring layer 342' and the wiring layer 343, and the connection portions 335. The pixel circuit 433 arranged in the semiconductor layer 303 and the memories 409 and 410 and the processing circuit 411 arranged in the semiconductor layer 304 are connected via through vias 336 provided in the semiconductor layer 303, the wiring patterns provided in a wiring layer 343' and the wiring layer 344, and the connection portions 337.

[0064] As described above, the connection portion 333 is arranged to connect the APD 201 arranged in the semiconductor layer 301 and the pixel circuit 432 arranged in the semiconductor layer 302. Although a manufacturing method will be described later, the connection portion 333 is a metal pattern that connects the wiring layer 341 formed to cover the semiconductor layer 301 and the wiring layer 342 formed to cover the semiconductor layer 302. As shown in FIGS. 2A and 6, the connection portion 333 is arranged to overlap each APD 201. As described above, the connection portion 335 is arranged to connect the pixel circuit 432 arranged in the semiconductor layer 302 and the pixel circuit 433 arranged in the semiconductor layer 303. The connection portion 335 is a metal pattern that connects the wiring layer 342' formed to cover the semiconductor layer 302 and the wiring layer 343 formed to cover the semiconductor layer 303. As shown in FIGS. 2B and 6, the connection portion 335 is arranged to overlap each pixel circuit 432. Each of the connection portions 333 and 335 is arranged to overlap the APD 201 and the pixel circuits 432 and 433 in the pixel formed by the APD 201 and the pixel circuits 432 and 433. Thus, miniaturization and integration of the pixels are implemented.

[0065] On the other hand, as shown in FIG. 6, the through vias 336 each for connecting the pixel circuit 433 arranged in the semiconductor layer 303 and the memories 409 and 410 and the processing circuit 411 arranged in the semiconductor layer 304 are arranged in a region not overlapping the APD array 401 in which the APDs 201 are arranged. If the through vias 336 extending through the semiconductor layer 303 are arranged in a region overlapping the APD array 401, a region necessary for not only the pixel circuits 433 but also the through vias 336 is necessary for the region overlapping the APD array 401. That is, it may be impossible to implement miniaturization of the pixel formed by the APD 201 and the pixel circuits 432 and 433. Therefore, the through vias 336 are arranged in a region not overlapping the APD array 401.

[0066] Furthermore, as described above, the connection portions 337 connected to the through vias 336 connect the signal transfer unit 406 arranged in the semiconductor layer 303 and the signal reception unit 407 arranged in the semiconductor layer 304. In other words, the connection portions 337 are arranged to connect the pixels each formed by the APD 201 and the pixel circuits 432 and 433 to the memories 409 and 410 and the processing circuit 411. The connection portion 337 is a metal pattern that connects the wiring layer 343' formed to cover the semiconductor layer 303 and the wiring layer 344 formed to cover the semiconductor layer 304. Since the through vias 336 are arranged in a region not overlapping the APD array 401, as described above, the connection portions 337 can also be arranged in a region not overlapping the APD array 401 in which the APDs 201 are arranged, as shown in FIG. 6. Therefore, as shown in FIGS. 2C and 6, it can be said that the signal transfer unit **406** and the signal reception unit **407** are arranged in a region not overlapping the APD array **401** in which the APDs **201** are arranged.

[0067] In this example, as shown in FIG. **6**, the semiconductor layer **302** may be thinner than the semiconductor layer **303**. In the semiconductor layer **302**, the through vias **334** the number of which is equal to the number of pixels each formed by the APD **201** and the pixel circuits **432** and **433** are formed. That is, the number of through vias **334** arranged in the semiconductor layer **302** may be larger than the number of through vias **334** arranged in the semiconductor layer **303**. Therefore, the semiconductor layer **302** may be thinner than the semiconductor layer **303** in consideration of the manufacturing step of the photoelectric conversion apparatus. By thinning the semiconductor layer **302**, the speed can be increased by improving the degree of integration of the pixels and reducing the parasitic capacitance in the through vias **334**.

[0068] The semiconductor layer **303** may be thinner than the semiconductor layer **304**. This is because the through vias **336** are formed in the semiconductor layer **303** but no through via is formed in the semiconductor layer **304**, as shown in FIG. **6**. By thinning the semiconductor layer **303**, the speed can be increased by improving the degree of integration of the pixels and reducing the parasitic capacitance in the through vias **336**. For example, the semiconductor layer **302** may be thinned to 1 μm or less. The semiconductor layer **303** may have a thickness of, for example, about 1 μm to 10 μm. On the other hand, the semiconductor layer **304** may have a thickness of, for example, about 50 μm to 100 μm. Furthermore, the semiconductor layer **304** need not be thinned to have a thickness of 725 μm or 775 μm. Furthermore, the semiconductor layer **301** may change in accordance with the wavelength of detected light and the like, and may have a thickness of, for example, about 2 μm to 10 μm.

[0069] As shown in FIG. **6**, the photoelectric conversion apparatus **100** may include one pad electrode **332** for supplying a common potential to the semiconductor layers **303** and **304**. The common potential may be a ground potential. The APD **201** and the pixel circuits **432** and **433** arranged in the semiconductor layers **301** to **303** can operate in accordance with a signal output from the scanning circuit **404** arranged in the semiconductor layer **303**. Thus, by supplying the common potential to the semiconductor layers **303** and **304**, a variation in power supply potential in each of the semiconductor layers **301** to **304** is suppressed, and for example, the output of the signal from the pixel is stabilized.

[0070] Furthermore, the width of the through via **336** extending through the semiconductor layer **303** may be larger than the width of the through via **334** extending through the semiconductor layer **302**. The width of the through via **334** or **336** may be the width of a thinnest portion of the through via **334** or **336** in a portion extending through the semiconductor layer **303** or **302**, in the sectional view shown in FIG. **6**. The through via **334** is provided for each pixel formed by the APD **201** and the pixel circuits **432** and **433**. It is difficult to arrange the pixel circuit **432** in the portion in which the through via **334** is arranged. To miniaturize the pixel and achieve integration, the through via **334** is formed to be smaller than the through via **336**. On the other hand, the number of through vias **336** is smaller than the number of through vias **334**, and the through vias **336** are arranged to overlap a region outside the APD array **401**. Thus, by increasing the width of the through via **336**, the resistance of the through via **336** is reduced, thereby improving the signal transfer ability from the pixel circuit **433** to the memories **409** and **410** and the processing circuit **411**. This increases the speed of the photoelectric conversion apparatus **100**.

[0071] As described above, the characteristic of the APD **201** may change along with a change of the temperature. On the other hand, the processing circuit **411** may execute calculation with a heavy load to improve the performance of the photoelectric conversion apparatus **100**, thereby increasing the heat generation amount. To cope with this, thermal design that suppresses heat generated in the semiconductor layer **304** from being transferred to the semiconductor layer **301** is required. To do this, in a planar view, the total area of the through vias **336** provided in the semiconductor layer **303** may be smaller than the total area of the through vias **334** provided in the semiconductor layer **302**. In consideration of the thermal design, in a planar view, the area of the through via **334** or **336** may be the area of a portion of the through via **334** or **336** whose section is smallest in a portion extending through the semiconductor layer **302** or **304**.

[0072] For the through vias **334** and **336**, a material having high thermal conductivity such as a metal can be used. Therefore, in a planar view, by making the total area of the through vias **336** smaller than the total area of the through vias **334**, heat generated in the semiconductor layer **304** is difficult to be transferred to the semiconductor layer **301**. The through via **336** can have a width larger than the width of the through via **334**, as described above. However, since the number of through vias **336** is smaller than the number of through vias **334**, the total area of the through vias **336** can be made small.

[0073] As described above, the through vias **336**, the connection portions **337**, the signal transfer unit **406**, and the signal reception unit **407** are arranged in a region not overlapping the APD array **401**. Thus, as compared to a case where the through vias **336**, the connection portions **337**, the signal transfer unit **406**, and the signal reception unit **407** are arranged to overlap the APD array **401**, it is possible to suppress heat transferred to the surface of the semiconductor layer **303** on the side of the semiconductor layer **302** from being transferred to the semiconductor layer **301**. While these components that easily transfer heat from the semiconductor layer **304** to the semiconductor layer **303** are arranged not to overlap the APD array **401**, the processing circuit **411** may be arranged to overlap the APD array **401**, as shown in FIGS. **2**A, **2**D, and **6**. This is because heat is mainly transferred from the semiconductor layer **304** to the semiconductor layer **303** via the through vias **336** and the connection portions **337**.

[0074] Although not shown in FIG. **6**, a phase synchronization circuit (phase locked loop (PLL) circuit) for controlling the operation timing of the overall photoelectric conversion apparatus **100** may be arranged in the semiconductor layer **304**. In the semiconductor layer **304**, a clock signal for controlling the overall photoelectric conversion apparatus **100** is generated and supplied to the semiconductor layers **301** to **304**. A frequency divider circuit and the like may be arranged in the semiconductor layers **301** to **303**. For example, a phase synchronization circuit (PLL circuit) may be arranged in each of the semiconductor layers **303** and **304**. One of the phase synchronization circuit arranged in the semiconductor layer **303** and the phase synchronization circuit arranged in the semiconductor layer **304** sends an original signal to the other, thereby generating a clock signal. The phase synchronization circuit arranged in the semiconductor layer **303** may generate a clock signal for controlling the respective components of the semiconductor layers **301** to **303**, and the phase synchronization circuit arranged in the semiconductor layer **304** generate a clock signal for controlling the respective components of the semiconductor layer **304**.

[0075] A manufacturing method of the photoelectric conversion apparatus **100** shown in FIGS. **5**B and **6** will be described next with reference to FIG. **7**. First, the semiconductor layers **301** to **304** respectively including the wiring layers **341** to **344** are prepared, as shown in FIG. **5**A. More specifically, the APDs **201** are formed in the semiconductor layer **301**, and the wiring layer **341** is formed to cover the semiconductor device forming region **340** in which the APDs **201** are arranged. The pixel circuits **432** and the like are formed in the semiconductor layer **302**, and the wiring layer **342** is formed to cover the semiconductor device forming region **340** in which the pixel circuits **432** are arranged. The pixel circuits **433** and the like are formed in the semiconductor layer **303**, and the wiring layer **343** is formed to cover the semiconductor device forming region **340** in which the pixel circuits **433** are arranged. The memories **409** and **410**, the processing circuit **411**, and the like are formed in the semiconductor layer **304**, and the wiring layer **344** is formed to cover the semiconductor device forming region **340** in which the memories **409** and **410** and the processing circuit **411** are arranged. The semiconductor layers **301** to **304** respectively including the wiring layers **341** to **344** may be manufactured in parallel or in an appropriate order. In each step shown in FIG. **7**, the necessary semiconductor layers **301** to **304** respectively including the wiring layers **341** to **344** need only be usable.

[0076] Next, as shown in a step **701**, the semiconductor layers **301** and **302** are bonded via the wiring layers **341** and **342**. The metal pattern arranged on the outermost surface of the wiring layer **341** and the metal pattern arranged on the outermost surface of the wiring layer **342** are bonded to form the above-described connection portions **333**.

[0077] After the semiconductor layers **301** and **302** are bonded, the semiconductor layer **302** is thinned from the opposite side of the semiconductor layer **301** and the through vias **334** and the like are formed in the semiconductor layer **302**, as shown in a step **702**. Furthermore, the wiring layer **342'** is formed to cover the surface of the semiconductor layer **302** on the opposite side of the semiconductor layer **301** (the surface on the opposite side of the semiconductor device forming region **340**).

[0078] Next, as shown in a step **703**, the semiconductor layers **302** and **303** are bonded via the wiring layers **342'** and **343**. The metal pattern arranged on the outermost surface of the wiring layer **342'** and the metal pattern arranged on the outermost surface of the wiring layer **343** are bonded to form the above-described connection portions **335**.

[0079] After the semiconductor layers **302** and **303** are bonded, the semiconductor layer **303** is thinned from the opposite side of the semiconductor layer **302** and the through vias **336** and the like are formed in the semiconductor layer **303**, as shown in a step **704**. Furthermore, the wiring layer **343'** is formed to cover the surface of the semiconductor layer **303** on the opposite side of the semiconductor layer **302** (the surface on the opposite side of the semiconductor device forming region **340**).

[0080] Next, as shown in a step **705**, the semiconductor layers **303** and **304** are bonded via the wiring layers **343'** and **344**. The metal pattern arranged on the outermost surface of the wiring layer **343'** and the metal pattern arranged on the outermost surface of the wiring layer **344** are bonded to form the above-described connection portions **337**.

[0081] After the semiconductor layers **303** and **304** are bonded, the semiconductor layer **301** is thinned from the opposite side of the semiconductor layer **302**, as shown in a step **706**. Next, the optical layer **345** such as a microlens is formed on the surface of the semiconductor layer **301** on the opposite side of the semiconductor layer **302** (the surface on the light incident side of the semiconductor layer **301**), thereby manufacturing the photoelectric conversion apparatus **100** shown in FIGS. **5**B and **6**.

[0082] FIG. **8** is a sectional view showing the more detailed configuration of the photoelectric conversion apparatus **100** shown in FIG. **5**C. In the configuration shown in FIGS. **5**B and **6**, in the semiconductor layer **302**, the semiconductor device forming region **340** in which the pixel circuits **432** and the like are arranged is arranged on the side of the semiconductor layer **301**. On the other hand, in the configuration shown in FIGS. **5**C and **8**, in the semiconductor layer **302**, the semiconductor device forming region **340** in which the pixel circuits **432** and the like are arranged is arranged on the side of the semiconductor layer **303**. The remaining components are the same as those described with reference to FIG. **6** and a description thereof will be omitted.

[0083] A manufacturing method of the photoelectric conversion apparatus **100** shown in FIGS. **5**C and **8** will be described next with reference to FIG. **9**. First, as described above, the semiconductor layers **301** to **304** respectively including the wiring layers **341** to **344** are prepared, as shown in FIG. **5**A. Next, as shown in a step **901**, the semiconductor layers **302** and **303** are bonded via the wiring layers **342** and **343**. The metal pattern arranged on the outermost surface of the wiring layer **342** and the metal pattern arranged on the outermost surface of the wiring layer **343** are bonded to form the above-described connection portions **335**.

[0084] After the semiconductor layers **302** and **303** are bonded, the semiconductor layer **302** is thinned from the opposite side of the semiconductor layer **303** and the through vias **334** and the like are formed in the semiconductor layer **302**, as shown in a step **902**. Furthermore, the wiring layer **342'** is formed to cover the surface of the semiconductor layer **302** on the opposite side of the semiconductor layer **303** (the surface on the opposite side of the semiconductor device forming region **340**).

[0085] Next, as shown in a step **903**, the semiconductor layers **301** and **302** are bonded via the wiring layers **341** and **342'**. The metal pattern arranged on the outermost surface of the wiring layer **341** and the metal pattern arranged on the outermost surface of the wiring layer **342'** are bonded to form the above-described connection portions **333**.

[0086] Steps after the semiconductor layers **301** and **302** are bonded and the semiconductor layers **301** to **303** are stacked may be the same as the steps **704** to **706** shown in FIG. **7**. Therefore, a description thereof will be omitted. By using these steps, the photoelectric conversion apparatus **100** shown in FIGS. **5**C and **8** is manufactured.

[0087] FIG. **10** is a sectional view showing a modification of the photoelectric conversion apparatus **100** shown in FIG. **6**. In the photoelectric conversion apparatus **100** shown in FIG. **10**, a configuration for connecting the pixel circuit **433** arranged in the semiconductor layer **303** to the memories **409** and **410** and the processing circuit **411** arranged in the semiconductor layer **304** is different from the configuration shown in FIG. **6**. In the configuration shown in FIG. **6**, the semiconductor layers **303** and **304** are connected via the through vias **336** provided in the semiconductor layer **303**, the wiring patterns provided in the wiring layers **343** and **344**, and the connection portions **337**. On the other hand, in the configuration shown in FIG. **10**, the through vias **336** extend through not only the semiconductor layer **303** but also the semiconductor layer **304** and are connected to connection portions **339** (to be also referred to as a wiring pattern hereinafter) provided in a wiring layer **344'** that covers the surface of the semiconductor layer **304** on the opposite side of the semiconductor layer **303**. Furthermore, the memories **409** and **410**, the processing circuit **411**, and the like arranged in the semiconductor layer **304** are connected to the connection portions **339** via the wiring pattern arranged in the wiring layer **344** and through vias **338** extending through the semiconductor layer **304**. That is, the semiconductor layers **303** and **304** are connected via the through vias **336** provided in the semiconductor layers **303** and **304**, the connection portions **339** provided in the wiring layer **344'**, the through vias **338** provided in the semiconductor layer **304**, and the wiring pattern provided in the wiring layer **344**. By connecting the semiconductor layers **303** and **304** in the semiconductor layer **304** on the opposite side of the semiconductor layer **303**, heat generated in the semiconductor layer **304** is difficult to be transferred to the semiconductor layer **301**. The remaining components may be the same as those described with reference to FIG. **6** and a description thereof will be omitted.

[0088] In the configuration shown in FIG. **10** as well, the width of the through via **336** extending through the semiconductor layer **303** may be larger than the width of the through via **334** extending through the semiconductor layer **302**. In a planar view, the total area of the through vias **336** provided in the semiconductor layer **303** may be smaller than the total area of the through vias **334** provided in the semiconductor layer **302**.

[0089] FIG. **11** is a sectional view showing a modification of the photoelectric conversion apparatus **100** shown in FIG. **10**. In the configuration shown in each of FIGS. **6**, **8**, and **10**, the pad electrodes **331** and **332** are exposed via openings formed in the semiconductor layer **301**. That is, each of the pad electrodes **331** and **332** is configured to be connectable from the light incident side of the photoelectric conversion apparatus **100** using a conductive wire or the like. On the other hand, in the configuration shown in FIG. **11**, the pad electrodes **331** and **332** are arranged on the surface of the wiring layer **344'** on the opposite side of the light incident side of the photoelectric conversion apparatus **100**. Therefore, the semiconductor region **312** in which the APDs **201** are formed is connected to the pad electrode **331** via the semiconductor regions **315** and **318**, the wiring patterns arranged in the wiring layers **341**, **342**, **342'**, and **343**, and the through vias **334** and **336**. The pad electrode **332** for supplying the common potential to the semiconductor layers **303** and **304** is directly or indirectly connected to the pixel circuit **433**, the memories **409** and **410**, the processing circuit **411**, and the like arranged in the semiconductor layers **303** and **304** via the through vias **336** and **338**. As shown in, for example, FIG. **11**, each of the pad electrodes **331** and **332** is connected to an electrode **351** of a wiring board **350** or an external apparatus of the photoelectric conversion apparatus **100** via a conductive member **352** such as a solder. This implements connection to an apparatus arranged outside the photoelectric conversion apparatus **100**. The pad electrodes **331** and **332** may be formed in a layer that is the same as or different from the layer of the above-described connection portions **339** in the wiring layer **344'**, as shown in FIG. **11**. The remaining components may be the same as those described with reference to FIG. **6** and a description thereof will be omitted.

[0090] In the configuration shown in each of FIGS. **10** and **11**, the pixel circuit **432** is arranged in the semiconductor layer **302** on the side of the semiconductor layer **301**. However, the present invention is not limited to this, and the pixel circuit **432** may be arranged in the semiconductor layer **302** on the side of the semiconductor layer **303**, similar to the configuration shown in FIGS. **5**C and **8**.

[0091] An application example of the above-described photoelectric conversion apparatus **100** will be described below.

[0092] FIG. **12** is a block diagram showing the schematic configuration of a photoelectric conversion system. The photoelectric conversion apparatus **100** described in each of the above-described embodiments is applicable to various kinds of photoelectric conversion systems. Examples of photoelectric conversion systems to which the photoelectric conversion apparatus is applicable are a digital still camera, a digital camcorder, a monitoring camera, a copying machine, a facsimile apparatus, a mobile phone, an in-vehicle camera, and an observation satellite. A camera module including an optical system such as a lens and an image capturing apparatus is also included in the photoelectric conversion systems. FIG. **12** exemplarily shows the block diagram of a digital still camera as an example of these.

[0093] A photoelectric conversion system **1000** exemplarily shown in FIG. **12** includes an image capturing apparatus **1004** as an example of the photoelectric conversion apparatus, a lens **1002** that forms an optical image of an object on the image capturing apparatus **1004**, an aperture **1003** configured to change the amount of light passing through the lens **1002**, and a barrier **1001** configured to protect the lens **1002**. The lens **1002** and the aperture **1003** form an optical system (optical apparatus) that condenses light to the image capturing apparatus **1004**. The image capturing apparatus **1004** is the photoelectric conversion apparatus **100** (image capturing apparatus) according to one of the above-described embodiments, and converts the optical image formed by the lens **1002** into an electrical signal.

[0094] The photoelectric conversion system **1000** also includes a signal processing unit **1007** that is an image generation unit configured to generate an image by processing an output signal output from the image capturing apparatus **1004**. The signal processing unit **1007** functions as a processing apparatus that performs an operation of performing various kinds of correction and compression as needed, thereby outputting image data. The signal processing unit **1007** may be formed on a semiconductor substrate on which the image capturing apparatus **1004** is provided or may be formed on a semiconductor substrate different from the image capturing apparatus **1004**. In addition, the image capturing apparatus **1004** and the signal processing unit **1007** may be formed on the same semiconductor substrate.

[0095] The photoelectric conversion system **1000** further includes a memory unit **1010** configured to temporarily store image data, and an external interface unit (external I/F unit) **1013** configured to communicate with an external computer or the like. Furthermore, the photoelectric conversion system **1000** includes a recording medium **1012** such as a semiconductor memory configured to record or read out image capturing data, and a recording medium control interface unit (recording medium control I/F unit) **1011** configured to perform record or readout for the recording medium **1012**. The recording medium control I/F unit **1011** and the recording medium **1012** can form a part of a recording apparatus. Note that the recording medium **1012** may be incorporated in the photoelectric conversion system **1000** or may be detachable.

[0096] Furthermore, the photoelectric conversion system **1000** includes a general control/arithmetic unit **1009** that controls various kinds of operations and the entire digital still camera, and a timing generation unit **1008** that outputs various kinds of timing signals to the image capturing apparatus **1004** and the signal processing unit **1007**. The general control/arithmetic unit **1009** and the timing generation unit **1008** can form a part of a control apparatus configured to control an operation of the photoelectric conversion system **1000**. In this example, the timing signal and the like may be input from the outside, and the photoelectric conversion system **1000** need only include at least the image capturing apparatus **1004**, and the signal processing unit **1007** that processes an output signal output from the image capturing apparatus **1004**.

[0097] The image capturing apparatus **1004** outputs an image capturing signal to the signal processing unit **1007**. The signal processing unit **1007** executes predetermined signal processing for the image capturing signal output from the image capturing apparatus **1004**, and outputs image data. The signal processing unit **1007** generates an image using the image capturing signal. Although not shown in FIG. **12**, a display apparatus such as a display for displaying the generated image may be arranged in the photoelectric conversion system **1000**. As described above, according to this embodiment, it is possible to implement the photoelectric conversion system **1000** to which the photoelectric conversion apparatus **100** (image capturing apparatus) according to one of the above-described embodiments is applied.

[0098] FIGS. **13**A and **13**B are views showing the configurations of a photoelectric conversion system **1300** and a moving body **1301**, respectively. FIG. **13**A shows an example of a photoelectric conversion system concerning an in-vehicle camera. The photoelectric conversion system **1300** includes an image capturing apparatus **1310**. The image capturing apparatus **1310** is the photoelectric conversion apparatus **100** (image capturing apparatus) described in one of the above-described embodiments. The photoelectric conversion system **1300** includes an image processing unit **1312** that performs image processing for a plurality of image data acquired by the image capturing apparatus **1310**. The photoelectric conversion system **1300** also includes a distance acquisition unit **1316** that calculates the distance up to a target object, and a collision determination unit **1318** that determines, based on the calculated distance, whether there is collision possibility. Here, the distance acquisition unit **1316** may acquire distance information up to a target object by using a Time of Flight (ToF) method, or may acquire distance information by using parallax information or the like. That is, the distance information is information concerning a parallax, a defocus amount, a distance up to a target object, and the like. The collision determination unit **1318** may determine collision possibility using one of the pieces of distance information. The distance acquisition unit **1316** may be implemented by exclusively designed hardware, or may be implemented by a software module. The distance acquisition unit **1316** may be implemented by a Field Programmable Gate Array (FPGA), an Application Specific Integrated Circuit (ASIC), or the like, or may be implemented by a combination of these.

[0099] The photoelectric conversion system **1300** is connected to a vehicle information acquisition apparatus **1320**, and can acquire vehicle information such as a vehicle speed, a yaw rate, and a steering angle. The photoelectric conversion system **1300** is also connected to an ECU **1330** that is a control apparatus configured to output a control signal for generating a braking force to the vehicle based on the determination result of the collision determination unit **1318**. Furthermore, the photoelectric conversion system **1300** is connected to an alarm apparatus **1340** that generates an alarm to the driver based on the determination result of the collision determination unit **1318**. For example, if collision possibility is high as the determination result of the collision determination unit **1318**, the ECU **1330** controls a driving apparatus (mechanical apparatus) **1360** to perform braking, releasing the accelerator pedal, or suppressing the engine output, thereby controlling the vehicle for avoiding collision and reducing damage. The alarm apparatus **1340** sounds an alarm, displays alarm information on the screen of a car navigation system or the like, or applies a vibration to the seat belt or a steering wheel, thereby making an alarm to the user.

[0100] In this embodiment, the periphery of the vehicle (moving body **1301**), for example, the front or rear side is captured by the photoelectric conversion system **1300**. FIG. **13**B shows the photoelectric conversion system when capturing the front side (image capturing range **1350**) of the vehicle. The vehicle information acquisition apparatus **1320** sends an instruction to the photoelectric conversion system **1300** or the image capturing apparatus **1310**. With this configuration, it is possible to further improve the accuracy of distance measurement.

[0101] An example in which control is executed so as not to collide with another vehicle has been explained above. The photoelectric conversion system **1300** can also be applied to control of performing automated driving following another vehicle or control of performing automated driving without deviating from a lane. Furthermore, the photoelectric conversion system **1300** can be applied not only to a vehicle such as an automobile but also to, for example, a moving body (moving apparatus) such as a ship, an airplane, or an industrial robot. The moving body includes one or both of a driving force generation unit that generates a driving force mainly used for moving the moving body and a rotating body mainly used for moving the moving body. The driving force generation unit can be an engine, a motor, or the like. The rotating body can be a tire, a wheel, a ship screw, an aircraft propeller, or the like. In addition, the photoelectric conversion system can be applied not only to a moving body but also to equipment that broadly uses object recognition, such as an intelligent transport system (ITS).

[0102] FIG. **14** is a block diagram showing an example of the configuration of a distance image sensor **1401** as the photoelectric conversion system. As shown in FIG. **14**, the distance image sensor **1401** includes an optical system **1402**, a photoelectric conversion apparatus **1403**, an image processing circuit **1404**, a monitor **1405**, and a memory **1406**. Then, the distance image sensor **1401** can receive light (modulated light or pulsed light) projected from a light source apparatus **1411** toward an object and reflected by the surface of the object, thereby acquiring a distance image corresponding to the distance up to the object.

[0103] The optical system **1402** is formed by including one or a plurality of lenses, and guides image light (incident light) from the object to the photoelectric conversion apparatus **1403** and forms an image on the light-receiving surface (sensor portion) of the photoelectric conversion apparatus **1403**.

[0104] As the photoelectric conversion apparatus **1403**, the photoelectric conversion apparatus **100** of each of the above-described embodiments is applied, and a distance signal indicating a distance obtained from a light reception signal output from the photoelectric conversion apparatus **1403** is supplied to the image processing circuit **1404**.

[0105] The image processing circuit **1404** performs image processing of creating a distance image based on the distance signal supplied from the photoelectric conversion apparatus **1403**. Then, the distance image (image data) obtained by the image processing is supplied to and displayed on the monitor **1405**, and supplied to and stored (recorded) in the memory **1406**.

[0106] The distance image sensor **1401** having such configuration can acquire, for example, a more correct distance image along with improvement in characteristic of pixels by applying the above-described photoelectric conversion apparatus **100**.

[0107] FIG. **15** is a view showing an example of the schematic configuration of an endoscopic surgery system **1250** as the photoelectric conversion system. FIG. **15** shows a state in which an operator (doctor) **1231** operates on a patient **1232** on a patient bed **1233** using the endoscopic surgery system **1250**. As shown in FIG. **15**, the endoscopic surgery system **1250** is formed from an endoscope **1200**, a surgical tool **1210**, and a cart **1234** on which various apparatuses for endoscopic surgery are mounted.

[0108] The endoscope **1200** includes a lens barrel **1201** including a region of a predetermined length from the distal end, which is inserted into the body cavity of the patient **1232**, and a camera head **1202** connected to the proximal end of the lens barrel **1201**. In the example shown in FIG. **15**, the endoscope **1200** formed as a so-called hard mirror including the hard lens barrel **1201** is shown but the endoscope **1200** may be formed as a so-called soft mirror including a soft lens barrel.

[0109] An opening in which an objective lens is fitted is provided at the distal end of the lens barrel **1201**. A light source apparatus **1203** is connected to the endoscope **1200**, and light generated by the light source apparatus **1203** is guided to the distal end of the lens barrel by a light guide extended inside the lens barrel **1201**, and is emitted to an observation target in the body cavity of the patient **1232** via the objective lens. Note that the endoscope **1200** may be a forward-viewing endoscope or may be a forward-oblique viewing endoscope or side-viewing endoscope.

[0110] An optical system and a photoelectric conversion apparatus are provided in the camera head **1202**, and reflected light (observation light) from the observation target is condensed by the optical system to the photoelectric conversion apparatus. The observation light is photoelectrically converted by the photoelectric conversion apparatus to generate an electrical signal corresponding to the observation light, that is, an image signal corresponding to an observation image. As the photoelectric conversion apparatus, the photoelectric conversion apparatus **100** (image capturing apparatus) described in each of the above-described embodiments can be used. The image signal is transmitted as RAW data to a Camera Control Unit (CCU) **1235**.

[0111] The CCU **1235** is formed by a Central Processing Unit (CPU), a Graphics Processing Unit (GPU), and the like, and comprehensively controls the operations of the endoscope **1200** and a display apparatus **1236**. Furthermore, the CCU **1235** receives an image signal from the camera head **1202**, and performs, for the image signal, various kinds of image processes such as development processing (demosaic processing) for displaying an image based on the image signal.

[0112] Under the control of the CCU **1235**, the display apparatus **1236** displays the image based on the image signal having undergone the image processing by the CCU **1235**.

[0113] The light source apparatus **1203** is formed from a light source such as a Light Emitting Diode (LED), and supplies, to the endoscope **1200**, irradiation light at the time of imaging an operation portion or the like.

[0114] An input apparatus **1237** is an input interface to the endoscopic surgery system **1250**. The user can input various kinds of information or instructions to the endoscopic surgery system **1250** via the input apparatus **1237**.

[0115] A treatment tool control apparatus **1238** controls driving of an energy treatment tool **1212** for ablation or incision of the tissue, sealing of a blood vessel, or the like.

[0116] The light source apparatus **1203** that supplies, to the endoscope **1200**, irradiation light at the time of imaging an operation portion can be formed from, for example, a white light source formed by an LED, a laser light source, or a combination thereof. If the white light source is formed by a combination of RGB laser light sources, it is possible to accurately control the output intensity and output timing of each color (each wavelength), and thus the light source apparatus **1203** can adjust the white balance of a captured image. In this case, the observation target is time-divisionally irradiated with laser beams from the RGB laser light sources, respectively, and driving of the image sensor of the camera head **1202** is controlled in synchronism with the irradiation timings, thereby making it possible to time-divisionally capture images respectively corresponding to R, G, and B. In this method, it is possible to obtain a color image without providing color filters in the image sensor.

[0117] Driving of the light source apparatus **1203** may be controlled to change the intensity of light to be output for every predetermined time. It is possible to time-divisionally acquire images by controlling driving of the image sensor of the camera head **1202** in synchronism with the timing of changing the intensity of the light, and combine the images, thereby generating an image of a high dynamic range without so-called shadow detail loss or highlight detail loss.

[0118] The light source apparatus **1203** may be configured to supply light in a predetermined wavelength band corresponding to special light observation. In special light observation, for example, the wavelength dependency of light absorption in the body tissue is used. More specifically, by performing irradiation with light in a narrow band, as compared with irradiation light (that is, white light) at the time of normal observation, predetermined tissue such as a blood vessel in the mucous membrane surface layer is captured with high contrast. Alternatively, in special light observation, fluorescence observation for obtaining an image by using fluorescence generated by performing irradiation with excitation light may be performed. In fluorescence observation, it is possible to, for example, irradiate body tissue with excitation light and observe fluorescence from the body tissue, or locally inject a reagent such as indocyanine green (ICG) to body tissue while irradiating the body tissue with excitation light corresponding to the fluorescence wavelength of the reagent, thereby obtaining a fluorescence image. The light source apparatus **1203** can be configured to supply narrow band light and/or excitation light corresponding to such special light observation.

[0119] FIGS. **16**A and **16**B each describe glasses **1600** (smartglasses) as the photoelectric conversion system. The glasses **1600** shown in FIG. **16**A include a photoelectric conversion apparatus **1602**. The photoelectric conversion apparatus **1602** is the photoelectric conversion apparatus **100** (image capturing apparatus) described in each of the above-described embodiments. A display apparatus including the light emitting apparatus such as an OLED or LED may be provided on the back surface side of a lens **1601**. One or a plurality of photoelectric conversion apparatuses **1602** may be provided. Alternatively, a plurality of kinds of photoelectric conversion apparatuses may be used in combination. The arrangement position of the photoelectric conversion apparatus **1602** is not limited to that shown in FIG. **16**A.

[0120] The glasses **1600** further include a control apparatus **1603**. The control apparatus **1603** functions as a power supply that supplies electric power to the photoelectric conversion apparatus **1602** and the above-described display apparatus. In addition, the control apparatus **1603** controls the operations of the photoelectric conversion apparatus **1602** and the display apparatus. An optical system configured to condense light to the photoelectric conversion apparatus **1602** is formed on the lens **1601**.

[0121] FIG. **16**B describes glasses **1610** (smartglasses) according to one application example. The glasses **1610** include a control apparatus **1612**, and a photoelectric conversion apparatus corresponding to the photoelectric conversion apparatus **1602** and a display apparatus are mounted on the control apparatus **1612**. The photoelectric conversion apparatus in the control apparatus **1612** and an optical system configured to project light emitted from the display apparatus are formed in a lens **1611**, and an image is projected to the lens **1611**. The control apparatus **1612** functions as a power supply that supplies electric power to the photoelectric conversion apparatus and the display apparatus, and controls the operations of the photoelectric conversion apparatus and the display apparatus. The control apparatus may include a line-of-sight detection unit that detects the line of sight of a wearer. The detection of a line of sight may be done using infrared rays. An infrared ray emitting unit emits infrared rays to an eyeball of the user who is gazing at a displayed image. An image capturing unit including a light receiving element detects reflected light of the emitted infrared rays from the eyeball, thereby obtaining a captured image of the eyeball. A reduction unit for reducing light from the infrared ray emitting unit to the display unit in a plan view is provided, thereby reducing deterioration of image quality.

[0122] The line of sight of the user to the displayed image is detected from the captured image of the eyeball obtained by capturing the infrared rays. An arbitrary known method can be applied to the line-of-sight detection using the captured image of the eyeball. As an example, a line-of-sight detection method based on a Purkinje image obtained by reflection of irradiation light by a cornea can be used.

[0123] More specifically, line-of-sight detection processing based on pupil center corneal reflection is performed. Using pupil center corneal reflection, a line-of-sight vector representing the direction (rotation angle) of the eyeball is calculated based on the image of the pupil and the Purkinje image included in the captured image of the eyeball, thereby detecting the line-of-sight of the user.

[0124] The display apparatus according to the embodiment can include a photoelectric conversion apparatus including a light receiving element, and control a displayed image of the display apparatus based on the line-of-sight information of the user from the photoelectric conversion apparatus.

[0125] More specifically, the display apparatus decides a first visual field region at which the user is gazing and a second visual field region other than the first visual field region based on the line-of-sight information. The first visual field region and the second visual field region may be decided by the control apparatus of the display apparatus, or those decided by an external control apparatus may be received. In the display region of the display apparatus, the display resolution of the first visual field region may be controlled to be higher than the display resolution of the second visual field region. That is, the resolution of the second visual field region may be lower than that of the first visual field region.

[0126] In addition, the display region includes a first display region and a second display region different from the first display region, and a region of higher priority may be decided from the first display region and the second display region based on line-of-sight information. The first visual field region and the second visual field region may be decided by the control apparatus of the display apparatus, or those decided by an external control apparatus may be received. The resolution of the region of higher priority may be controlled to be higher than the resolution of the region other than the region of higher priority. That is, the resolution of the region of relatively low priority may be low.

[0127] Note that AI may be used to decide the first visual field region or the region of higher priority. The AI may be a model configured to estimate the angle of the line of sight and the distance to a target object ahead the line of sight from the image of the eyeball using the image of the eyeball and the direction of actual viewing of the eyeball in the image as supervised data. The AI program may be held by the display apparatus, the photoelectric conversion apparatus, or an external apparatus. If the external apparatus holds the AI program, it is transmitted to the display apparatus via communication.

[0128] If display control is performed based on line-of-sight detection, this can be applied to smartglasses further including a photoelectric conversion apparatus configured to capture the image of the outside. The smartglasses can display the captured outside image information in real time.

[0129] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0130] This application claims the benefit of Japanese Patent Application No. 2024-034179, filed Mar. 6, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A back-illuminated type photoelectric conversion apparatus in which a first semiconductor layer including an APD array in which a plurality of avalanche photodiodes (APDs) are arranged, a second semiconductor layer including a plurality of first pixel circuits respectively for the plurality of APDs, a third semiconductor layer including a plurality of second pixel circuits respectively for the plurality of first pixel circuits, and a fourth semiconductor layer including a processing circuit configured to process signals output from the plurality of second pixel circuits are stacked, wherein

the second semiconductor layer is arranged between the first semiconductor layer and the fourth semiconductor layer,

the third semiconductor layer is arranged between the second semiconductor layer and the fourth semiconductor layer, and

a through via extending through the third semiconductor layer is arranged in a region not overlapping the APD array.

2. The apparatus according to claim **1**, wherein the plurality of second pixel circuits are arranged on a surface of the third semiconductor layer on a side of the second semiconductor layer.

3. The apparatus according to claim **1**, wherein with respect to the plurality of first pixel circuits, one first pixel circuit is arranged in correspondence with each of the plurality of APDs or one first pixel circuit is arranged in correspondence with one group constituted by a predetermined number of APDs among the plurality of APDs.

4. The apparatus according to claim **1**, wherein the processing circuit is arranged on a surface of the fourth semiconductor layer on a side of the third semiconductor layer.

5. The apparatus according to claim **1**, wherein a memory configured to store the signals output from the plurality of second pixel circuits is arranged in the fourth semiconductor layer.

6. The apparatus according to claim **5**, wherein

the memory includes a first memory and a second memory configured to store the signals output from the plurality of second pixel circuits or signals obtained by processing the signals output from the plurality of second pixel circuits, and

the processing circuit is arranged between the first memory and the second memory.

7. The apparatus according to claim **1**, further comprising:

one pad electrode configured to supply a common potential to the third semiconductor layer and the fourth semiconductor layer.

8. The apparatus according to claim **7**, wherein the common potential is a ground potential.

9. The apparatus according to claim **1**, wherein a pad electrode exposed via an opening formed in the first semiconductor layer and connected to the plurality of APDs is arranged between the first semiconductor layer and the second semiconductor layer.

10. The apparatus according to claim **1**, wherein the second semiconductor layer is thinner than the third semiconductor layer.

11. The apparatus according to claim **1**, wherein the third semiconductor layer is thinner than the fourth semiconductor layer.

12. The apparatus according to claim **1**, wherein a width of the through via extending through the third semiconductor layer is larger than a width of a through via extending through the second semiconductor layer.

13. The apparatus according to claim **1**, wherein in a planar view, a total area of the through via provided in the third semiconductor layer is smaller than a total area of the through via provided in the second semiconductor layer.

14. The apparatus according to claim **1**, wherein

a signal transfer unit configured to transfer the signals output from the plurality of second pixel circuits to the fourth semiconductor layer is arranged in the third semiconductor layer, and

the signal transfer unit is arranged in a region not overlapping the APD array.

15. The apparatus according to claim **14**, wherein in a planar view, the signal transfer unit is arranged along a long side of the APD array.

16. The apparatus according to claim **1**, wherein the processing circuit is arranged to overlap the APD array.

17. The apparatus according to claim **1**, wherein a phase synchronization circuit configured to control an operation timing of the overall photoelectric conversion apparatus is arranged in the fourth semiconductor layer.

18. The apparatus according to claim **1**, wherein a phase synchronization circuit is arranged in each of the third semiconductor layer and the fourth semiconductor layer.

19. A photoelectric conversion system comprising:

a photoelectric conversion apparatus defined in claim **1**; and

a signal processing unit configured to generate an image using a signal output by the photoelectric conversion apparatus.

20. A moving body that includes a photoelectric conversion apparatus defined in claim **1**, comprising:

a control unit configured to control movement of the moving body using a signal output by the photoelectric conversion apparatus.

21. Equipment that comprises a photoelectric conversion apparatus defined in claim **1**, further comprising at least one of:

an optical apparatus corresponding to the photoelectric conversion apparatus;

a control apparatus configured to control the photoelectric conversion apparatus;

a processing apparatus configured to process a signal output from the photoelectric conversion apparatus;

a display apparatus configured to display information obtained by the photoelectric conversion apparatus;

a storage apparatus configured to store information obtained by the photoelectric conversion apparatus; and

a mechanical apparatus configured to operate based on information obtained by the photoelectric conversion apparatus.

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